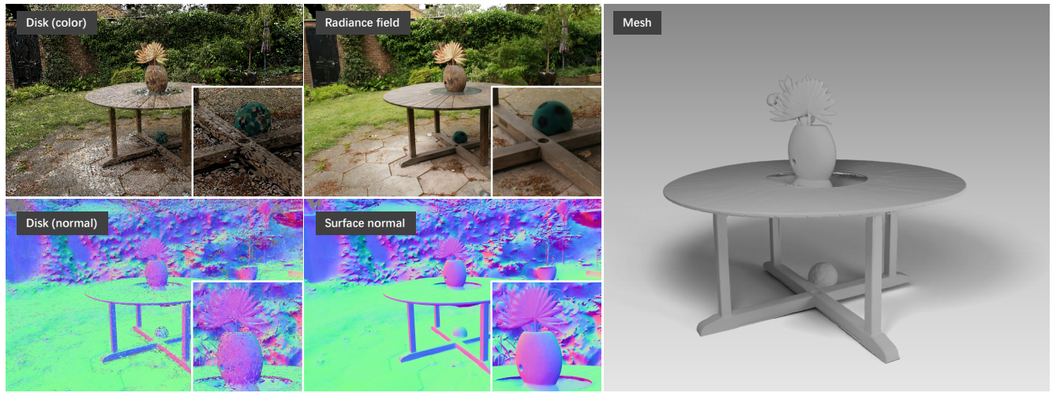
本文档由 funstory.ai 的开源 PDF 翻译库 BabelDOC v0.6.4 (https://github.com/funstory-ai/BabelDOC) 翻译，本仓库正在积极的建设当中，欢迎 star 和关注。

arXiv:2403.17888v3 [cs.CV] 22 Feb 2025

2D Gaussian Splatting for Geometrically Accurate Radiance Fields

BINBIN HUANG, ShanghaiTech University, China
ZEHAO YU, University of Tübingen Tübingen AI Center, Germany
ANPEI CHEN, University of Tübingen Tübingen AI Center, Germany
ANDREAS GEIGER, University of Tübingen Tübingen AI Center, Germany
SHENGHUA GAO, ShanghaiTech University, China
https://surfsplatting.github.io



(a) 2D disks as surface elements (b) 2D Gaussian splatting (c) Meshing

Fig. 1. Our method, *2DGS*, (a) optimizes a set of 2D oriented disks to represent and reconstruct a complex real-world scene from multi-view RGB images. These optimized 2D disks are tightly aligned to the surfaces. (b) With 2D Gaussian splatting, we allow real-time rendering of high quality novel view images with view consistent normals and depth maps. (c) Finally, our method provides detailed and noise-free triangle mesh reconstruction from the optimized 2D disks.

3D Gaussian Splatting (3DGS) has recently revolutionized radiance field reconstruction, achieving high quality novel view synthesis and fast rendering speed. However, 3DGS fails to accurately represent surfaces due to the multi-view inconsistent nature of 3D Gaussians. We present 2D Gaussian Splatting (2DGS), a novel approach to model and reconstruct geometrically accurate radiance fields from multi-view images. Our key idea is to collapse the 3D volume into a set of 2D oriented planar Gaussian disks. Unlike 3D Gaussians, 2D Gaussians provide view-consistent geometry while modeling surfaces intrinsically. To accurately recover thin surfaces and achieve stable optimization, we introduce a perspective-accurate 2D splatting process utilizing ray-splat intersection and rasterization. Additionally, we incorporate depth distortion and normal consistency terms to further enhance the quality of the reconstructions. We demonstrate that our differentiable renderer allows for noise-free and detailed geometry reconstruction while maintaining competitive appearance quality, fast training speed, and real-time rendering.

CCS Concepts: • **Computing methodologies** → *Reconstruction*; **Rendering**; *Machine learning approaches.*

Additional Key Words and Phrases: Novel View Synthesis, Radiance Fields, Surface Splatting, Surface Reconstruction

ACM Reference Format:
Binbin Huang, Zehao Yu, Anpei Chen, Andreas Geiger, and Shenghua Gao. 2024. 2D Gaussian Splatting for Geometrically Accurate Radiance Fields. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers '24 (SIGGRAPH Conference Papers '24), July 27-August 1, 2024, Denver, CO, USA.* ACM, New York, NY, USA, 13 pages. https://doi.org/10.1145/3641519.3657428

Authors' Contact Information: Binbin Huang, huangbb@shanghaitech.edu.cn, ShanghaiTech University, Shanghai, China; Zehao Yu, zehao.yu@uni-tuebingen.de, University of Tübingen and Tübingen AI Center, Tübingen, Germany; Anpei Chen, anpei.chen@uni-tuebingen.de, University of Tübingen and Tübingen AI Center, Tübingen, Germany; Andreas Geiger, a.geiger@uni-tuebingen.de, University of Tübingen and Tübingen AI Center, Tübingen, Germany; Shenghua Gao, gaoshh@shanghaitech.edu.cn, ShanghaiTech University, Shanghai, China.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
SIGGRAPH Conference Papers '24, July 27-August 1, 2024, Denver, CO, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0525-0/24/07
https://doi.org/10.1145/3641519.3657428

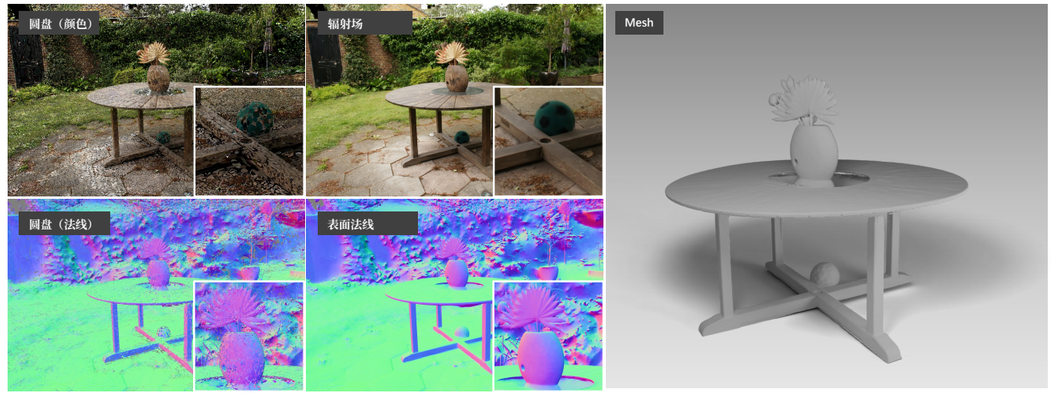
1 INTRODUCTION

Photorealistic novel view synthesis (NVS) and accurate geometry reconstruction stand as pivotal long-term objectives in computer graphics and vision. Recently, 3D Gaussian Splatting (3DGS) [Kerbl

基于几何精度的辐射场二维高斯喷溅

黄斌斌，上海科技大学，中国余哲豪，德国蒂宾根大学蒂宾根人工智能中心陈安培，德国蒂宾根大学蒂宾根人工智能中心安德烈亚斯·盖格，德国蒂宾根大学蒂宾根人工智能中心高胜华，上海科技大学，中国https://surfsplatting.github.io

arXiv:2403.17888v3 [cs.CV] 22 Feb 2025



(a) 2D圆盘作为表面元素 (b) 2D高斯喷溅 (c) 网格化

图1。我们的方法、*2DGS*，(a) 优化一组二维定向圆盘，以表示并从多视角RGB图像中重建复杂的真实世界场景。这些优化的二维圆盘与表面紧密对齐。(b) 通过二维高斯喷溅，我们允许实时渲染具有视图一致法线和深度图的高质量新视角图像。(c) 最后，我们的方法从优化的二维圆盘提供详细且无噪声的三角形网格重建。

3D高斯喷溅 (3DGS) 近年来革新了辐射场重建，实现了高质量的新视角合成和快速渲染速度。然而，由于3D高斯的多视图不一致性，3DGS无法准确表示表面。我们提出了2D高斯喷溅 (2DGS)，一种从多视图图像中建模和重建几何精确辐射场的新方法。我们的核心思想是将3D体积压缩为一组二维定向平面高斯圆盘。与3D高斯不同，2D高斯在建模表面时提供视图一致性，而3D高斯则不具备这一特性。为了准确恢复薄表面并实现稳定优化，我们引入了一种透视精确的2D喷溅过程，该过程利用光线喷溅交点和光栅化。此外，我们结合了深度失真和法线一致性项，以进一步提高重建质量。我们证明了我们的可微渲染器能够实现无噪声和精细的几何重建，同时保持有竞争力的外观质量、快速训练速度和实时渲染。

CCS概念：• **计算方法** → **重建**；**渲染**；机器学习方法。

其他关键词和短语：新视角合成、辐射场、表面喷溅、表面重建

ACM参考格式：
黄斌斌，余哲豪，陈安培，安德烈亚斯·盖格和 高胜华. 2024. 2D高斯喷溅用于几何精确的辐射场. 在 计算机图形学与交互技术特别兴趣小组会议论文*'24 (SIGGRAPH*会议论文*'24)*，*2024*年*7*月*27*日至*8*月*1*日，丹佛，科罗拉多州，美国. ACM, 纽约, 纽约, 美国, 13页.
https://doi.org/10.1145/3641519.3657428

作者联系方式：黄斌斌，huangbb@shanghaitech.edu.cn，上海交通大学，上海，中国；余哲豪，zehao.yu@uni-tuebingen.de，蒂宾根大学和蒂宾根AI中心，蒂宾根，德国；陈安培，anpei.chen@uni-tuebingen.de，蒂宾根大学和蒂宾根AI中心，蒂宾根，德国；安德烈亚斯·盖格，a.geiger@uni-tuebingen.de，蒂宾根大学和蒂宾根AI中心，蒂宾根，德国；高胜华，gaoshh@shanghaitech.edu.cn，上海Tech大学，上海，中国。

允许为个人或课堂教学目的免费制作本文全部或部分内容的数字或纸质副本，但不得以盈利或商业优势为目的制作或分发副本，并且副本须包含本声明和第一页上的完整引用。本作品第三方组件的版权必须得到尊重。对于所有其他用途，请联系所有者/作者。*SIGGRAPH*会议论文*'24*，*2024*年*7*月*27*日至*8*月*1*日，丹佛，科罗拉多州，美国© 2024 版权由所有者/作者持有。ACM ISBN 979-8-4007-0525-0/24/07 https://doi.org/10.1145/3641519.3657428

1 引言

照片级真实感新视角合成(NVS)和精确几何重建是计算机图形学和视觉领域中的关键长期目标。最近，3D高斯喷溅 (3DGS) [Kerbl

et al. 2023] has emerged as an appealing alternative to implicit [Barron et al. 2022a; Mildenhall et al. 2020] and feature grid-based representations [Barron et al. 2023; Müller et al. 2022] in NVS, due to its real-time photorealistic NVS results at high resolutions. Rapidly evolving, 3DGS has been quickly extended with respect to multiple domains, including anti-aliasing rendering [Yu et al. 2024], material modeling [Jiang et al. 2023; Shi et al. 2023], dynamic scene reconstruction [Yan et al. 2023], and animatable avatar creation [Qian et al. 2023; Zielonka et al. 2023]. Nevertheless, it falls short in capturing intricate geometry since the volumetric 3D Gaussian, which models the complete angular radiance, conflicts with the thin nature of surfaces.

On the other hand, earlier works [Pfister et al. 2000; Zwicker et al. 2001a,b] have shown surfels (surface elements) to be an effective representation of complex geometry. Surfels approximate the object surface locally with shape and shade attributes and can be derived from known geometry. They are widely used in SLAM [Whelan et al. 2016] and other robotics tasks [Schöps et al. 2019] as an efficient geometry representation. Subsequent advancements [Yifan et al. 2019] have incorporated surfels into a differentiable framework. However, these methods typically require ground truth (GT) geometry, depth sensor data, or operate under constrained scenarios with known lighting.

Inspired by these works, we propose 2D Gaussian Splatting for 3D scene reconstruction and novel view synthesis that combines the benefits of both worlds, while overcoming their limitations. Unlike 3DGS, our approach represents a 3D scene with 2D Gaussian primitives, each defining an oriented elliptical disk. The significant advantage of 2D Gaussian over its 3D counterpart lies in the accurate geometry representation during rendering. Specifically, 3DGS evaluates a Gaussian's value at the intersection between a pixel ray and a 3D Gaussian [Keselman and Hebert 2022, 2023], which leads to inconsistency depth when rendered from different viewpoints. In contrast, our method utilizes explicit ray-splat intersection, resulting in a perspective correct splatting, as illustrated in Figure 2, which in turn significantly improves reconstruction quality. Furthermore, the inherent surface normals in 2D Gaussian primitives enable direct surface regularization through normal constraints. In contrast with surfels-based models [Pfister et al. 2000; Yifan et al. 2019; Zwicker et al. 2001a], our 2D Gaussians can be recovered from unknown geometry with gradient-based optimization.

While our 2D Gaussian approach excels in geometric modeling, optimizing solely with photometric losses can lead to noisy reconstructions, due to the inherently unconstrained nature of 3D reconstruction tasks, as noted in [Barron et al. 2022b; Yu et al. 2022b; Zhang et al. 2020]. To enhance reconstructions and achieve smoother surfaces, we introduce two regularization terms: depth distortion and normal consistency. The depth distortion term concentrates 2D primitives distributed within a tight range along the ray, addressing the rendering process's limitation where the distance between Gaussians is ignored. The normal consistency term minimizes discrepancies between the rendered normal map and the gradient of the rendered depth, ensuring alignment between the geometries defined by depth and normals. Employing these regularizations in combination with our 2D Gaussian model enables us to extract highly accurate surface meshes, as demonstrated in Figure 1.

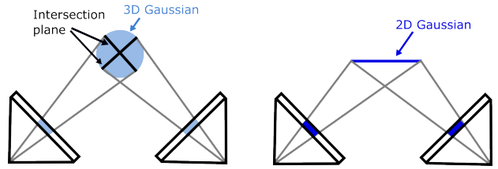


Fig. 2. Comparison of 3DGS and 2DGS. 3DGS utilizes different intersection planes for value evaluation when viewing from different viewpoints, resulting in inconsistency. Our 2DGS provides multi-view consistent value evaluations.

In summary, we make the following contributions:

- We present a highly efficient differentiable 2D Gaussian renderer, enabling perspective-correct splatting by leveraging 2D surface modeling, ray-splat intersection, and volumetric integration.
- We introduce two regularization losses for improved and noise-free surface reconstruction.
- Our approach achieves state-of-the-art geometry reconstruction and NVS results compared to other explicit representations.

2 RELATED WORK

2.1 Novel view synthesis

Significant advancements have been achieved in NVS, particularly since the introduction of Neural Radiance Fields (NeRF) [Mildenhall et al. 2021]. NeRF employs a multi-layer perceptron (MLP) to represent geometry and view-dependent appearance, optimized via volume rendering to deliver exceptional rendering quality. Post-NeRF developments have further enhanced its capabilities. For instance, Mip-NeRF [Barron et al. 2021] and subsequent works [Barron et al. 2022a, 2023; Hu et al. 2023] tackle NeRF's aliasing issues. Additionally, the rendering efficiency of NeRF has seen substantial improvements through techniques such as distillation [Reiser et al. 2021; Yu et al. 2021] and baking [Chen et al. 2023a; Hedman et al. 2021; Reiser et al. 2023; Yariv et al. 2023]. Moreover, the training and representational power of NeRF have been enhanced using feature-grid based scene representations [Chen et al. 2022, 2023c; Fridovich-Keil et al. 2022; Liu et al. 2020; Müller et al. 2022; Sun et al. 2022a].

Recently, 3D Gaussian Splatting (3DGS) [Kerbl et al. 2023] has emerged, demonstrating impressive real-time NVS results. This method has been quickly extended to multiple domains [Xie et al. 2023; Yan et al. 2023; Yu et al. 2024; Zielonka et al. 2023]. In this work, we propose to "flatten" 3D Gaussians to 2D Gaussian primitives to better align their shape with the object surface. Combined with two novel regularization losses, our approach reconstructs surfaces more accurately than 3DGS while preserving its high-quality and real-time rendering capabilities.

2.2 3D reconstruction

3D Reconstruction from multi-view images has been a long-standing goal in computer vision. Multi-view stereo based methods [Schönberger et al. 2016; Yao et al. 2018; Yu and Gao 2020] rely on a modular

et al. 2023] 已成为一种有吸引力的替代方案，用于新视角合成(NVS)，它取代了隐式 [Bar- ron等人 2022a; Mildenhall等人 2020] 以及基于特征网格的表示 [Barron等人 2023; Müller等人 2022]，由于其能够在高分辨率下实现实时照片级真实感的NVS结果。快速发展的3DGS已被迅速扩展到多个领域，包括抗锯齿渲染 [Yu 等人 2024]，材质建模 [Jiang 等人 2023; Shi 等人 2023]，动态场景重建 [Yan等人 2023]，以及可动画头像创建 [Qian等人 2023; Zielonka等人 2023]。然而，它在捕捉复杂几何方面存在不足，因为体素化3D高斯（它建模了完整的角辐射）与表面的薄性质相冲突。

另一方面，早期工作 [Pfister等人2000年; Zwicker等人 2001a,b] 已经表明，表面元素（surfels）是复杂几何形状的有效表示方法。表面元素通过形状和阴影属性在局部近似物体表面，并且可以从已知几何形状中推导出来。它们被广泛应用于SLAM [Whelan等人 2016] 以及其他机器人任务 [Schöps等人 2019]，作为高效的几何表示方法。后续的进展 [Yifan等人 2019] 将表面元素整合到了可微分的框架中。然而，这些方法通常需要真值（GT）几何形状、深度传感器数据，或在已知光照条件的约束场景下运行。

受这些工作的启发，我们提出了用于3D场景重建和新视角合成的二维高斯基元喷溅（2D Gaussian Splatting），它结合了这两个领域的优点，同时克服了它们的局限性。与3DGS不同，我们的方法使用二维高斯基元表示3D场景，每个基元定义一个定向的椭圆盘。二维高斯基元相对于其三维对应物的显著优势在于渲染过程中的精确几何表示。具体来说，3DGS在像素光线与三维高斯基元 [Keselman和Hebert 2022, 2023], 的交点处评估高斯基元的值，这会导致从不同视点渲染时出现不一致的深度。相比之下，我们的方法利用显式的光线喷溅交点，实现了透视正确的喷溅，如图2所示，这反过来显著提高了重建质量。此外，二维高斯基元中固有的表面法线使得可以通过法线约束直接进行表面正则化。与基于表面元素的模型 [Pfister等人2000年; Yifan等人2019年; Zwicker等人2001a],不同，我们的二维高斯基元可以通过基于梯度的优化从未知几何形状中恢复。

尽管我们的2D高斯方法在几何建模方面表现出色，但仅通过光度损失进行优化可能会导致重建结果存在噪声，这是由于3D重建任务固有的无约束性质所致，正如 [Barron等人 2022b；Yu等人2022b；Zhanget等人 2020]所指出的。为了提升重建效果并实现更平滑的表面，我们引入了两个正则化项：深度失真和法向一致性。深度失真项集中分布在沿光线紧密范围内的2D基元，解决了渲染过程中高斯之间距离被忽略的限制。法向一致性项最小化渲染法线图与渲染深度梯度之间的差异，确保深度和法线定义的几何体之间保持一致。结合我们的2D高斯模型，采用这些正则化方法使我们能够提取出高度精确的表面网格，如图1所示。

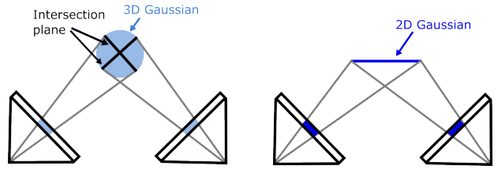


图2. 3DGS与2DGS的比较。3DGS在从不同视点观察时使用不同的交点平面进行值评估，导致不一致性。我们的2DGS提供多视图一致性的值评估。

总之，我们做出了以下贡献：

- 我们提出了一种高效的微分式2D高斯渲染器，通过利用2D表面建模、光线喷溅交点和体积积分实现透视校正喷溅。
- 我们引入了两种正则化损失，以实现更好的无噪声表面重建。
- 与其它显式表示相比，我们的方法在几何重建和NVS结果上达到了当前最佳水平。

2 相关工作

2.1 新视角合成

在NVS领域取得了显著进展，特别是自引入神经辐射场（NeRF）[Milden-hall等人 2021]以来。NeRF采用多层感知器（MLP）来表示几何形状和视点相关的外观，通过体积渲染进行优化，以提供卓越的渲染质量。NeRF后续的发展进一步增强了其功能。例如，Mip-NeRF [Barron等人 2021]及后续工作 [Barron等人2022a, 2023; Hu等人 2023] 解决了NeRF的走样问题。此外，通过蒸馏 [Reiser等人2021; Yu等人 2021] 和烘焙 [Chen等人2023a; Hedman等人2021; Reiser等人2023; Yariv等人 2023]等技术，NeRF的渲染效率得到了大幅提升。此外，基于特征网格的场景表示 [Chen等人2022, 2023c; Fridovich-Keil等人2022; Liu等人2020; Müller等人2022; Sun等人2022a]被用于增强NeRF的训练和表示能力。

最近，3D高斯喷溅 (3DGS) [Kerbl等人 2023] 已出现，展示了令人印象深刻的实时NVS结果。该方法已迅速扩展到多个领域 [Xie等人2023；Yan等人2023；Yu等人2024；Zielonka等人 2023]。在这项工作中，我们提出将3D高斯“扁平化”为2D高斯基元，以更好地使它们的形状与物体表面对齐。结合两种新的正则化损失，我们的方法在比3DGS更精确地重建表面的同时，保留了其高质量和实时渲染能力。

2.2 3D重建

3D重建从多视图图像开始一直是计算机视觉中的一个长期目标。基于多视图立体的方法 [Schön-berger等人2016年；姚等人2018年；余和高 2020] 依赖于模块化

pipeline that involves feature matching, depth prediction, and fusion. In contrast, recent neural approaches [Niemeyer et al. 2020; Yariv et al. 2020] represent surface implicitly via an MLP [Mescheder et al. 2019; Park et al. 2019] , extracting surfaces post-training via the Marching Cube algorithm. Further advancements [Oechsle et al. 2021; Wang et al. 2021; Yariv et al. 2021] integrated implicit surfaces with volume rendering, achieving detailed surface reconstructions from RGB images. These methods have been extended to large-scale reconstructions via additional regularization [Li et al. 2023; Yu et al. 2022a,b], and efficient reconstruction for objects [Wang et al. 2023]. Despite these impressive developments, efficient large-scale scene reconstruction remains a challenge. For instance, Neuralangelo [Li et al. 2023] requires 128 GPU hours for reconstructing a single scene from the Tanks and Temples Dataset [Knapitsch et al. 2017]. In this work, we introduce 2D Gaussian splatting, a method that significantly accelerates the reconstruction process. It achieves similar or slightly better results compared to previous implicit neural surface representations, while being an order of magnitude faster.

2.3 Differentiable Point-based Graphics

Differentiable point-based rendering [Aliev et al. 2020; Insafutdinov and Dosovitskiy 2018; Rückert et al. 2022; Wiles et al. 2020; Yifan et al. 2019] has been explored extensively due to its efficiency and flexibility in representing intricate structures. Notably, NPBG [Aliev et al. 2020] rasterizes point cloud features onto an image plane, subsequently utilizing a convolutional neural network for RGB image prediction. DSS [Yifan et al. 2019] focuses on optimizing oriented point clouds from multi-view images under known lighting conditions. Pulsar [Lassner and Zollhofer 2021] introduces a tile-based acceleration structure for more efficient rasterization. More recently, 3DGS [Kerbl et al. 2023] optimizes anisotropic 3D Gaussian primitives, demonstrating real-time photorealistic NVS results. Despite these advances, using point-based representations from unconstrained multi-view images remains challenging. In this paper, we demonstrate detailed surface reconstruction using 2D Gaussian primitives. We also highlight the critical role of additional regularization losses in optimization, showcasing their significant impact on the quality of the reconstruction.

2.4 Concurrent works

Since 3DGS [Kerbl et al. 2023] was introduced, it has been rapidly adapted across multiple domains. We now review the closest work in inverse rendering. These work [Gao et al. 2023; Jiang et al. 2023; Liang et al. 2023; Shi et al. 2023] extend 3DGS by modeling normals as additional attributes of 3D Gaussian primitives. Our approach, in contrast, inherently defines normals by representing the tangent space of the 3D surface using 2D Gaussian primitives, aligning them more closely with the underlying geometry. Additionally, the aforementioned works predominantly focus on estimating the material properties of the scene and evaluating their results for relighting tasks. Notably, none of these works specifically target surface reconstruction, the primary focus of our work.

We also highlight the distinctions between our method and concurrent works SuGaR [Guédon and Lepetit 2023] and NeuSG [Chen et al. 2023b]. Unlike SuGaR, which approximates 2D Gaussians with 3D Gaussians, our method directly employs 2D Gaussians, simplifying the process and enhancing the resulting geometry without additional mesh refinement. NeuSG optimizes 3D Gaussian primitives and an implicit SDF network jointly and extracts the surface from the SDF network, while our approach leverages 2D Gaussian primitives for surface approximation, offering a faster and conceptually simpler solution.

3 3D GAUSSIAN SPLATTING

Kerbl et al. [Kerbl et al. 2023] propose to represent 3D scenes with 3D Gaussian primitives and render images using differentiable volume splatting. Specifically, 3DGS explicitly parameterizes Gaussian primitives via 3D covariance matrix Σ and their location $\mathbf{p}_k$:

$$\mathcal{G}(\mathbf{p}) = \exp(-\frac{1}{2}(\mathbf{p}-\mathbf{p}_k)^\top \Sigma^{-1}(\mathbf{p}-\mathbf{p}_k)) \tag{1}$$

where the covariance matrix $\Sigma = \mathbf{RSS}^\top\mathbf{R}^\top$ is factorized into a scaling matrix $\mathbf{S}$ and a rotation matrix $\mathbf{R}$. To render an image, the 3D Gaussian is transformed into the camera coordinates with world-to-camera transform matrix $\mathbf{W}$ and projected to image plane via a local affine transformation $\mathbf{J}$ [Zwicker et al. 2001a]:

$$\Sigma' = \mathbf{JW}\Sigma\mathbf{W}^\top\mathbf{J}^\top \tag{2}$$

By skipping the third row and column of Σ', we obtain a 2D Gaussian $\mathcal{G}^{2D}$ with covariance matrix Σ^{2D}. Next, 3DGS [Kerbl et al. 2023] employs volumetric alpha blending to integrate alpha-weighted appearance from front to back:

$$\mathbf{c}(\mathbf{x}) = \sum_{k=1}^{K} \mathbf{c}_k\, \alpha_k\, \mathcal{G}_k^{2D}(\mathbf{x}) \prod_{j=1}^{k-1}(1-\alpha_j\, \mathcal{G}_j^{2D}(\mathbf{x})) \tag{3}$$

where k is the index of the Gaussian primitives, α_k denotes the alpha values and $\mathbf{c}_k$ is the view-dependent appearance. The attributes of 3D Gaussian primitives are optimized using a photometric loss.

Challenges in Surface Reconstruction. Reconstructing surfaces using 3D Gaussian modeling and splatting faces several challenges. First, the volumetric radiance representation of 3D Gaussians conflicts with the thin nature of surfaces. Second, 3DGS does not natively model surface normals, essential for high-quality surface reconstruction. Third, the rasterization process in 3DGS lacks multi-view consistency, leading to varied 2D intersection planes for different viewpoints [Keselman and Hebert 2023], as illustrated in Figure 2 (a). Additionally, using an affine matrix for transforming a 3D Gaussian into ray space only yields accurate projections near the center, compromising on perspective accuracy around surrounding regions [Zwicker et al. 2004]. Therefore, it often results in noisy reconstructions, as shown in Figure 5.

4 2D GAUSSIAN SPLATTING

To accurately reconstruct geometry while maintaining high-quality novel view synthesis, we present differentiable 2D Gaussian splatting (2DGS).

基于多视图图像的3D重建一直是计算机视觉中的一个长期目标。基于多视图立体的方法 [Schön- berger等人2016年；Yao等人2018年；Yu和高 2020] 依赖于一个包含特征匹配、深度预测和融合的模块化流程。相比之下，最近的神经方法 [Niemeyer等人2020年；Yariv等人 2020] 通过MLP隐式表示表面 [Mescheder等人2019年；Park等人 2019] , 在训练后通过Marching Cube算法提取表面。进一步的进展 [Oechsle等人2021年；Wang等人2021年；Yariv等人 2021] 将隐式表面与体积渲染集成，从RGB图像中实现了详细的表面重建。这些方法通过额外的正则化 [Li等人2023年；Yu等人2022a,b], 扩展到大规模重建，以及Wang等人 2023]对物体的高效重建。尽管取得了这些令人印象深刻的进展，但高效的大规模场景重建仍然是一个挑战。例如，Neuralangelo [Li等人 2023] 从坦克与寺庙数据集 [Knapitsch等人 2017]重建单个场景需要128个GPU小时。在这项工作中，我们引入了二维高斯splatting，这是一种显著加速重建过程的方法。它在结果上与之前的隐式神经表面表示相似或略好，同时速度提高了一个数量级。

2.3 可微点云图形

可微点云渲染 [Aliev等人2020；Insafutdinov和Dosovitskiy2018；Rückert等人2022；Wiles等人2020；Yifan等人 2019] 已得到广泛探索，因其高效性和灵活性，在表示复杂结构方面表现出色。值得注意的是，NPBG [Aliev等人 2020] 将点云特征光栅化到图像平面，随后利用卷积神经网络进行RGB图像预测。DSS [Yifan等人 2019] 专注于在已知光照条件下优化多视图图像中的定向点云。Pulsar [Lassner和Zollhofer 2021] 引入了一种基于瓦片的加速结构，以实现更高效的光栅化。最近，3DGS [Kerbl等人 2023] 优化了各向异性3D高斯基元，展示了实时照片级真实感NVS结果。尽管取得了这些进展，但从无约束多视图图像中获取基于点的表示仍然具有挑战性。在本文中，我们展示了使用2D高斯基元进行详细表面重建。我们还强调了优化中附加正则化损失的关键作用，展示了它们对重建质量的重要影响。

2.4 并行工作

自3D高斯喷溅3DGS [由Kerbl等人 2023] 提出以来，它已被迅速应用于多个领域。我们现在回顾逆向渲染中最接近的工作。这些工作 [Gao等人2023；Jiang等人2023；Liang等人2023；Shi等人 2023] 通过将法线建模为3D高斯基元的附加属性来扩展3DGS。相比之下，我们的方法通过使用2D高斯基元表示3D表面的切空间来固有地定义法线，更紧密地与底层几何一致。此外，上述工作主要关注估计场景的材料属性，并评估其结果用于重光照任务。值得注意的是，这些工作中的没有一项专门针对表面重建——这是我们工作的主要焦点。

我们还强调了我们的方法与同时期工作SuGaR [Guédon和Lepetit 2023] 以及NeuSG [Chen等人2023b]之间的区别。与SuGaR不同，后者使用3D高斯来近似2D高斯基元，我们的方法直接采用2D高斯基元，简化了流程，并在无需额外网格细化的情况下提升了最终几何效果。NeuSG联合优化3D高斯基元和隐式SDF网络，并从SDF网络中提取表面，而我们的方法利用2D高斯基元进行表面近似，提供了一种更快且概念上更简单的解决方案。

3 3D高斯喷溅

Kerbl等人 [Kerbl等人 2023] 提出使用3D高斯基元表示3D场景，并使用可微体积喷溅渲染图像。具体来说，3DGS通过3D协方差矩阵 Σ 及其位置 $\mathbf{p}_k$显式地参数化高斯基元：

$$\mathcal{G}(\mathbf{p}) = \exp(-\frac{1}{2}(\mathbf{p}-\mathbf{p}_k)^\top \Sigma^{-1}(\mathbf{p}-\mathbf{p}_k)) \tag{1}$$

在协方差矩阵 $\Sigma = \mathbf{RSS}^\top\mathbf{R}^\top$ 被分解为一个缩放矩阵 $\mathbf{S}$ 和一个旋转矩阵 $\mathbf{R}$ 的地方。为了渲染图像，3D高斯通过世界到相机的变换矩阵 $\mathbf{W}$ 转换到相机坐标系，并通过局部仿射变换 $\mathbf{J}$ [Zwicker等人2001a] 投影到图像平面：

$$\Sigma' = \mathbf{JW}\Sigma\mathbf{W}^\top\mathbf{J}^\top \tag{2}$$

通过跳过 Σ' 的第三行和列，我们得到一个具有协方差矩阵 Σ^{2D} 的二维高斯 G^{2D}。接下来，3DGS [Kerbl等人 2023] 采用体积alpha混合来从前到后集成alpha加权的外观：

$$\mathbf{c}(\mathbf{x}) = \sum_{k=1}^{K} \mathbf{c}_k\, \alpha_k\, \mathcal{G}_k^{2D}(\mathbf{x}) \prod_{j=1}^{k-1}(1-\alpha_j\, \mathcal{G}_j^{2D}(\mathbf{x})) \tag{3}$$

其中 k 是高斯原语的索引， α_k 表示alpha值，$\mathbf{c}_k$ 是视点相关的外观。3D高斯原语的属性使用光度损失进行优化。

表面重建中的挑战。 使用3D高斯建模和喷溅重建表面存在几个挑战。首先，3D高斯的体积辐射表示与表面的薄性质相冲突。其次，3DGS 本身不自然地建模表面法线，这对于高质量表面重建至关重要。第三，3DGS 中的光栅化过程缺乏多视图一致性，导致不同视点的二维交线平面不同 [Keselman和Hebert 2023], 如图2 (a) 所示。此外，使用仿射矩阵将3D高斯转换为射线空间仅能产生中心附近的准确投影，而周围区域的透视精度会受到影响 [Zwicker等人 2004]。因此，它通常会导致噪声较大的重建，如图5所示。

4 2D 高斯投射

为了在保持高质量新视角合成的同时精确重建几何结构，我们提出了可微分的二维高斯喷溅（2DGS）。

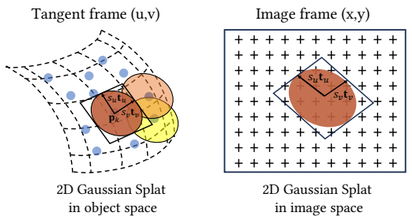


Fig. 3. Illustration of 2D Gaussian Splatting. 2D Gaussian Splats are elliptical disks characterized by a center point $\mathbf{p}_k$, tangential vectors $\mathbf{t}_u$ and $\mathbf{t}_v$, and two scaling factors (s_u and s_v) control the variance. Their elliptical projections are sampled through the ray-splat intersection (Section 4.2) and accumulated via alpha-blending in image space. 2DGS reconstructs surface attributes such as colors, depths, and normals through gradient descent.

4.1 Modeling

Unlike 3DGS [Kerbl et al. 2023], which models the entire angular radiance in a blob, we simplify the 3-dimensional modeling by adopting "flat" 2D Gaussians embedded in 3D space. With 2D Gaussian modeling, the primitive distributes densities within a planar disk, defining the normal as the direction of the steepest change of density. This feature enables better alignment with thin surfaces. While previous methods [Kopanas et al. 2021; Yifan et al. 2019] also utilize 2D Gaussians for geometry reconstruction, they require a dense point cloud or ground-truth normals as input. By contrast, we simultaneously reconstruct the appearance and geometry given only a sparse calibration point cloud and photometric supervision.

As illustrated in Figure 3, our 2D splat is characterized by its central point $\mathbf{p}_k$, two principal tangential vectors $\mathbf{t}_u$ and $\mathbf{t}_v$, and a scaling vector $\mathbf{S} = (s_u, s_v)$ that controls the variances of the 2D Gaussian. Notice that the primitive normal is defined by two orthogonal tangential vectors $\mathbf{t}_w = \mathbf{t}_u \times \mathbf{t}_v$. We can arrange the orientation into a 3×3 rotation matrix $\mathbf{R} = [\mathbf{t}_u, \mathbf{t}_v, \mathbf{t}_w]$ and the scaling factors into a 3×3 diagonal matrix $\mathbf{S}$ whose last entry is zero.

A 2D Gaussian is therefore defined in a local tangent plane in world space, which is parameterized:

$$P(u,v) = \mathbf{p}_k + s_u\mathbf{t}_u u + s_v\mathbf{t}_v v = \mathbf{H}(u,v,1,1)^{\mathrm{T}} \tag{4}$$

$$\text{where } \mathbf{H} = \begin{bmatrix} s_u\mathbf{t}_u & s_v\mathbf{t}_v & \mathbf{0} & \mathbf{p}_k \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{RS} & \mathbf{p}_k \\ \mathbf{0} & 1 \end{bmatrix} \tag{5}$$

where $\mathbf{H} \in 4 \times 4$ is a homogeneous transformation matrix representing the geometry of the 2D Gaussian. For the point $\mathbf{u} = (u, v)$ in uv space, its 2D Gaussian value can then be evaluated by standard Gaussian

$$\mathcal{G}(\mathbf{u}) = \exp\left(-\frac{u^2+v^2}{2}\right) \tag{6}$$

The center $\mathbf{p}_k$, scaling (s_u, s_v), and the rotation $(\mathbf{t}_u, \mathbf{t}_v)$ are learnable parameters. Following 3DGS [Kerbl et al. 2023], each 2D Gaussian primitive has opacity α and view-dependent appearance c parameterized with spherical harmonics.

4.2 Splatting

One common strategy for rendering 2D Gaussians is to project the 2D Gaussian primitives onto the image space using the affine approximation of the perspective projection [Zwicker et al. 2001a,b]. However, as noted in [Zwicker et al. 2004], this projection is only accurate at the center of the Gaussian and has increasing approximation error with increased distance to the center. To address this issue, Zwicker et al. proposed a formulation based on homogeneous coordinates. Specifically, projecting the 2D splat onto an image plane can be described by a general 2D-to-2D mapping in homogeneous coordinates. Let $\mathbf{W} \in 4 \times 4$ be the combined transformation matrix from world space to screen space. The screen space points are hence obtained by

$$\mathbf{x} = (xz, yz, z, z)^{\mathrm{T}} = \mathbf{W}P(u,v) = \mathbf{WH}(u,v,1,1)^{\mathrm{T}} \tag{7}$$

where $\mathbf{x}$ represents a homogeneous ray emitted from the camera and passing through pixel (x, y) and intersecting the splat at depth z. To rasterize a 2D Gaussian, Zwicker et al. proposed to project its conic into the screen space with an implicit method using $\mathbf{M} = (\mathbf{WH})^{-1}$. However, the inverse transformation introduces numerical instability, especially when the splat degenerates into a line segment (i.e., if it is viewed from the side). To address this issue, previous surface splatting rendering methods discard such ill-conditioned transformations using a predefined threshold [Zwicker et al. 2004]. However, such a scheme poses challenges within a differentiable rendering framework, as thresholding can lead to unstable optimization. To address this problem, we utilize an explicit ray-splat intersection inspired by [Sigg et al. 2006].

Ray-splat Intersection. We efficiently locate the ray-splat intersections by finding the intersection of three non-parallel planes, a method originally designed for specialized hardware [Weyrich et al. 2007]. Given an image coordinate $\mathbf{x} = (x, y)$, we parameterize the ray of a pixel in the projective space as the intersection of two orthogonal planes: the x-plane and the y-plane. Specifically, the x-plane is defined by a normal vector $(-1, 0, 0)$ and an offset x. The x-plane can be represented as a 4D homogeneous plane $\mathbf{h}_x = (-1, 0, 0, x)^{\mathrm{T}}$. Similarly, the y-plane is $\mathbf{h}_y = (0, -1, 0, y)^{\mathrm{T}}$. Thus, the ray $\mathbf{x} = (x, y)$ is determined by the intersection of the two planes.

Next, we transform both planes into the local coordinates of the 2D Gaussian primitives, the uv-coordinate system. Note that transforming points on a plane using a transformation matrix $\mathbf{M}$ is equivalent to transforming homogeneous plane parameters using the inverse transpose $\mathbf{M}^{-\mathrm{T}}$ [Blinn 1977]. Therefore, applying $\mathbf{M} = (\mathbf{WH})^{-1}$ is equivalent to $(\mathbf{WH})^{\mathrm{T}}$, eliminating explicit matrix inversion and yielding:

$$\mathbf{h}_u = (\mathbf{WH})^{\mathrm{T}}\mathbf{h}_x \quad \mathbf{h}_v = (\mathbf{WH})^{\mathrm{T}}\mathbf{h}_y \tag{8}$$

As introduced in Section 4.1, points on the 2D Gaussian plane are represented as $(u, v, 1, 1)$. At the same time, the intersection point should fall in the transformed x-plane and y-plane. Thus,

$$\mathbf{h}_u \cdot (u,v,1,1)^{\mathrm{T}} = \mathbf{h}_v \cdot (u,v,1,1)^{\mathrm{T}} = 0 \tag{9}$$

This leads to an efficient solution for the intersection point $\mathbf{u}(\mathbf{x})$:

$$u(\mathbf{x}) = \frac{\mathbf{h}_u^2\mathbf{h}_v^4 - \mathbf{h}_u^4\mathbf{h}_v^2}{\mathbf{h}_u^1\mathbf{h}_v^2 - \mathbf{h}_u^2\mathbf{h}_v^1} \qquad v(\mathbf{x}) = \frac{\mathbf{h}_u^4\mathbf{h}_v^1 - \mathbf{h}_u^1\mathbf{h}_v^4}{\mathbf{h}_u^1\mathbf{h}_v^2 - \mathbf{h}_u^2\mathbf{h}_v^1} \tag{10}$$

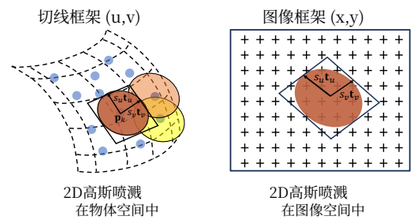


图3. 2D高斯喷溅的示意图。2D高斯喷溅是具有中心点 $\mathbf{p}_k$、切向向量 $\mathbf{t}_u$ 和 $\mathbf{t}_v$，以及两个控制方差的缩放因子（s_u 和 s_v）的椭圆盘。它们的椭圆投影通过光线喷溅交点（第4.2节）采样，并在图像空间中通过alpha混合累积。2DGS通过梯度下降重建颜色、深度和法线等表面属性。

4.1 建模

与3DGS [Kerbl等人 2023], 不同，后者将整个角辐射度建模在一个体积内，我们通过采用嵌入在三维空间中的"平面"二维高斯来简化三维建模。通过二维高斯建模，该原语在平面圆盘内分布密度，将法线定义为密度变化最快的方向。这一特性使得与薄表面更好地对齐。虽然之前的方法 [Kopanas等人 2021；Yifan等人 2019] 也利用二维高斯进行几何重建，但它们需要密集点云或真实法线作为输入。相比之下，我们仅使用稀疏标定点云和光度监督，即可同时重建外观和几何结构。

如图3所示，我们的2D喷溅以其中心点 $\mathbf{p}_k$、两个主切向向量 $\mathbf{t}_u$ 和 $\mathbf{t}_v$，以及控制2D高斯方差的缩放向量 $\mathbf{S} = (s_u, s_v)$ 为特征。注意，原始法线由两个正交切向向量 $\mathbf{t}_w = \mathbf{t}_u \times \mathbf{t}_v$定义。我们可以将方向排列成一个 3×3 旋转矩阵 $\mathbf{R} = [\mathbf{t}_u, \mathbf{t}_v, \mathbf{t}_w]$，并将缩放因子排列成一个 3×3 对角矩阵 $\mathbf{S}$，其最后一个元素为零。

因此，2D高斯在局部切平面内定义于世界空间中，该平面被参数化：

$$P(u,v) = \mathbf{p}_k + s_u\mathbf{t}_u u + s_v\mathbf{t}_v v = \mathbf{H}(u,v,1,1)^{\mathrm{T}} \tag{4}$$

$$\text{where } \mathbf{H} = \begin{bmatrix} s_u\mathbf{t}_u & s_v\mathbf{t}_v & \mathbf{0} & \mathbf{p}_k \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{RS} & \mathbf{p}_k \\ \mathbf{0} & 1 \end{bmatrix} \tag{5}$$

其中 $\mathbf{H} \in 4 \times 4$ 是表示2D高斯几何的齐次变换矩阵。对于点 $\mathbf{u} = (u, v)$ 在uv 空间中，其2D高斯值可以通过标准高斯方法进行评估

$$\mathcal{G}(\mathbf{u}) = \exp\left(-\frac{u^2+v^2}{2}\right) \tag{6}$$

中心 $\mathbf{p}_k$、缩放 (s_u, s_v) 以及旋转 $(\mathbf{t}_u, \mathbf{t}_v)$ 都是可学习参数。遵循 3DGS [Kerbl等人 2023],，每个 2D高斯基元都具有不透明度 α 和依赖于视角的外观 c 参数，这些参数使用球谐函数进行参数化。

4.2 喷溅

渲染 2D高斯基元的一种常见策略是使用透视投影的仿射近似 [Zwicker等人 2001a,b] 将 2D高斯基元投影到图像空间。然而，如 [Zwicker等人 2004], 所述，这种投影仅在高斯基元的中心是准确的，并且随着与中心距离的增加，近似误差会越来越大。为了解决这个问题，Zwicker等人提出了一种基于齐次坐标的公式。具体来说，将 2Dsplat 投影到图像平面可以由齐次坐标中的通用 2D到2D映射来描述。令 $\mathbf{W} \in 4 \times 4$ 为从世界空间到屏幕空间的组合变换矩阵。因此，屏幕空间点是通过

$$\mathbf{x} = (xz, yz, z, z)^{\mathrm{T}} = \mathbf{W}P(u,v) = \mathbf{WH}(u,v,1,1)^{\mathrm{T}} \tag{7}$$

其中 $\mathbf{x}$ 表示从相机发射并通过像素 (x, y) 与喷溅在深度 z 处相交的齐次射线。为了光栅化2D高斯，Zwicker等人提出使用 $\mathbf{M} = (\mathbf{WH})^{-1}$ 通过隐式方法将其锥形投影到屏幕空间。然而，逆变换会引入数值不稳定性，尤其是在喷溅退化为线段时（即从侧面观察时）。为了解决这个问题，以前表面喷溅渲染方法使用预定义的阈值 [Zwicker等人 2004] 来丢弃这种病态变换。然而，这种方案在可微渲染框架中会带来挑战，因为阈值化会导致优化不稳定。为了解决这个问题，我们利用了受 [Sigg等人 2006] 启发的显式光线喷溅交点方法。

光线喷溅交点。我们通过寻找三个非平行平面的交点来高效定位光线喷溅交点，该方法最初是为专用硬件 [Weyrich et al. 2007] 设计的。给定图像坐标 $\mathbf{x} = (x, y)$, 我们将投影空间中像素的光线参数化为两个正交平面的交点：x平面和y平面。具体来说，x平面由法向量 $(-1, 0, 0)$ 和一个偏移量 x 定义。x平面可以用4D齐次平面 $\mathbf{h}_x = (-1, 0, 0, x)^{\mathrm{T}}$ 表示。类似地，y平面是 $\mathbf{h}^y = (0, -1, 0, y)^{\mathrm{T}}$。因此，光线 $\mathbf{x} = (x, y)$ 由两个平面的交点确定。

接下来，我们将两个平面转换到2D高斯基元的局部坐标，即 uv-坐标系。请注意，使用变换矩阵$\mathbf{M}$变换平面上的点等同于使用逆转置$\mathbf{M}^{-\mathrm{T}}$[Blinn 1977]变换齐次平面参数。因此，应用$\mathbf{M} = (\mathbf{WH})^{-1}$ 等同于 $(\mathbf{WH})^{\mathrm{T}}$，消除了显式的矩阵求逆，并得到：

$$\mathbf{h}_u = (\mathbf{WH})^{\mathrm{T}}\mathbf{h}_x \quad \mathbf{h}_v = (\mathbf{WH})^{\mathrm{T}}\mathbf{h}_y \tag{8}$$

如第4.1节所述，2D高斯平面上的点表示为$(u, v, 1, 1)$。同时，交点应落在变换后的 x-平面和 y-平面上。因此，

$$\mathbf{h}_u \cdot (u,v,1,1)^{\mathrm{T}} = \mathbf{h}_v \cdot (u,v,1,1)^{\mathrm{T}} = 0 \tag{9}$$

这导致了对交点$\mathbf{u}(\mathbf{x})$的求解方法有效：

$$u(\mathbf{x}) = \frac{\mathbf{h}_u^2\mathbf{h}_v^4 - \mathbf{h}_u^4\mathbf{h}_v^2}{\mathbf{h}_u^1\mathbf{h}_v^2 - \mathbf{h}_u^2\mathbf{h}_v^1} \qquad v(\mathbf{x}) = \frac{\mathbf{h}_u^4\mathbf{h}_v^1 - \mathbf{h}_u^1\mathbf{h}_v^4}{\mathbf{h}_u^1\mathbf{h}_v^2 - \mathbf{h}_u^2\mathbf{h}_v^1} \tag{10}$$

where $\mathbf{h}_u^i, \mathbf{h}_v^i$ are the i-th parameter of the 4D plane. Note that $\mathbf{h}_u^3$ and $\mathbf{h}_v^3$ are always zero according to Eq. 5. Once we obtain the local coordinates (u, v), we can calculate the depth z of the intersected points using Eq. 7 and evaluate the Gaussian value with Eq. 6.

Degenerate Solutions. When a 2D Gaussian is observed from a slanted viewpoint, it degenerates to a line in screen space. Therefore, it might be missed during rasterization. To deal with these cases and stabilize optimization, we employ the object-space low-pass filter introduced in [Botsch et al. 2005]:

$$\hat{\mathcal{G}}(\mathbf{x}) = \max\left\{\mathcal{G}(\mathbf{u}(\mathbf{x})), \mathcal{G}(\frac{\mathbf{x}-\mathbf{c}}{\sigma})\right\} \tag{11}$$

where $\mathbf{u}(\mathbf{x})$ is given by Eq. 10 and $\mathbf{c}$ is the projection of center $\mathbf{p}_k$. Intuitively, $\hat{\mathcal{G}}(\mathbf{x})$ is lower-bounded by a fixed screen-space Gaussian low-pass filter with center $\mathbf{c}$ and radius σ. In our experiments, we set $\sigma = \sqrt{2}/2$ to ensure sufficient pixels are used during rendering.

Rasterization. We follow a similar rasterization process as in 3DGS [Kerbl et al. 2023]. First, a screen space bounding box is computed for each Gaussian primitive. Then, 2D Gaussians are sorted based on the depth of their center and organized into tiles based on their bounding boxes. Finally, volumetric alpha blending is used to integrate alpha-weighted appearance from front to back:

$$\mathbf{c}(\mathbf{x}) = \sum_{i=1} \mathbf{c}_i \, \alpha_i \, \hat{\mathcal{G}}_i(\mathbf{u}(\mathbf{x})) \prod_{j=1}^{i-1} (1 - \alpha_j \, \hat{\mathcal{G}}_j(\mathbf{u}(\mathbf{x}))) \tag{12}$$

The iterative process is terminated when the accumulated opacity reaches saturation.

5 TRAINING

Our 2D Gaussian method, while effective in geometric modeling, can result in noisy reconstructions when optimized only with photometric losses, a challenge inherent to 3D reconstruction tasks [Barron et al. 2022b; Yu et al. 2022b; Zhang et al. 2020]. To mitigate this issue and improve the geometry reconstruction, we introduce two regularization terms: depth distortion and normal consistency.

Depth Distortion. Different from NeRF, 3DGS's volume rendering doesn't consider the distance between intersected Gaussian primitives. Therefore, spreading out Gaussians might result in a similar color and depth rendering. This is different from surface rendering, where rays intersect the first visible surface exactly once. To mitigate this issue, we take inspiration from Mip-NeRF360 [Barron et al. 2022a] and propose a depth distortion loss to concentrate the weight distribution along the rays by minimizing the distance between the ray-splat intersections:

$$\mathcal{L}_d = \sum_{i,j} \omega_i \omega_j |z_i - z_j| \tag{13}$$

where $\omega_i = \alpha_i \hat{\mathcal{G}}_i(\mathbf{u}(\mathbf{x})) \prod_{j=1}^{i-1}(1 - \alpha_j \hat{\mathcal{G}}_j(\mathbf{u}(\mathbf{x})))$ is the blending weight of the i–th intersection and z_i is the depth of the intersection points. Unlike the distortion loss in Mip-NeRF360, where z_i is the distance between sampled points and is not optimized, our approach directly encourages the concentration of the splats by adjusting the intersection depth z_i. Note that we implement this regularization term efficiently with CUDA in a manner similar to [Sun et al. 2022b].

Normal Consistency. As our representation is based on 2D Gaussian surface elements, we must ensure that all 2D splats are locally aligned with the actual surfaces. In the context of volume rendering where multiple semi-transparent surfels may exist along the ray, we consider the actual surface at the median point of intersection $\mathbf{p}_s$, where the accumulated opacity reaches 0.5. We then align the splats' normal with the gradients of the depth maps as follows:

$$\mathcal{L}_n = \sum_i \omega_i (1 - \mathbf{n}_i^{\mathrm{T}} \mathbf{N}) \tag{14}$$

where i indexes over intersected splats along the ray, ω denotes the blending weight of the intersection point, $\mathbf{n}_i$ represents the normal of the splat that is oriented towards the camera, and $\mathbf{N}$ is the normal estimated by the gradient of the depth map. Specifically, $\mathbf{N}$ is computed with finite differences from nearby depth points as follows:

$$\mathbf{N}(x, y) = \frac{\nabla_x \mathbf{p}_s \times \nabla_y \mathbf{p}_s}{|\nabla_x \mathbf{p}_s \times \nabla_y \mathbf{p}_s|} \tag{15}$$

By aligning the splat normal with the estimated surface normal, we ensure that 2D splats locally approximate the actual object surface.

Final Loss. Finally, we optimize our model from an initial sparse point cloud using a set of posed images. We minimize the following loss function:

$$\mathcal{L} = \mathcal{L}_c + \alpha \mathcal{L}_d + \beta \mathcal{L}_n \tag{16}$$

where $\mathcal{L}c$ is an RGB reconstruction loss combining $\mathcal{L}_1$ with the D-SSIM term from [Kerbl et al. 2023], while $\mathcal{L}_d$ and $\mathcal{L}_n$ are regularization terms. We set $\alpha = 1000$ for bounded scenes, $\alpha = 100$ for unbounded scenes, and $\beta = 0.05$ for all scenes.

6 EXPERIMENTS

We now present evaluations of our 2D Gaussian Splatting reconstruction method, including appearance and geometry comparison with previous state-of-the-art implicit and explicit approaches. We then analyze the contribution of the proposed components.

6.1 Implementation

We implement our 2D Gaussian Splatting with custom CUDA kernels, building upon the framework of 3DGS [Kerbl et al. 2023]. We extend the renderer to output depth distortion maps, depth maps and normal maps for regularizations (See detailed computations in Appendices A and B of the supplemental material). During training, we increase the number of 2D Gaussian primitives following the adaptive control strategy in 3DGS. Since our method does not directly rely on the gradient of the projected 2D center, we hence project the gradient of 3D center $\mathbf{p}_k$ onto the screen space as an approximation. Similarly, we employ a gradient threshold of 0.0002 and remove splats with opacity lower than 0.05 every 3000 step. We conduct all the experiments on a single GTX RTX3090 GPU.

Mesh Extraction. To extract meshes from reconstructed 2D splats, we render depth maps of the training views using the depth value of the splats projected to the pixels and utilize truncated signed distance fusion (TSDF) to fuse the reconstruction depth maps, using Open3D [Zhou et al. 2018]. We set the voxel size to 0.004 and the truncated threshold to 0.02 during TSDF fusion. We also extend the

其中 $\mathbf{h}_u^i, \mathbf{h}_v^i$ 是4D平面的第 i个参数。请注意，根据公式5，$\mathbf{h}_u^3$ 和 $\mathbf{h}_v^3$ 始终为零。一旦我们获得局部坐标(u, v)，我们就可以使用公式7计算交点的深度 z，并使用公式6评估高斯值。

退化解。 当从倾斜视角观察二维高斯时，它在屏幕空间中退化为一条线。因此，在光栅化过程中可能会被遗漏。为了处理这些情况并稳定优化，我们采用了 [Botsch等人 2005] 中引入的对象空间低通滤波器：

$$\hat{\mathcal{G}}(\mathbf{x}) = \max\left\{\mathcal{G}(\mathbf{u}(\mathbf{x})), \mathcal{G}(\frac{\mathbf{x}-\mathbf{c}}{\sigma})\right\} \tag{11}$$

其中 $\mathbf{u}(\mathbf{x})$ 由公式 10 给出，$\mathbf{c}$ 是中心 $\mathbf{p}_k$ 的投影。直观地，$\mathrm{G}(\mathbf{x})$ 被一个以 $\mathbf{c}$ 为中心、半径 σ 的固定屏幕空间高斯低通滤波器下界。在我们的实验中，我们设置 $\sigma = \sqrt{2}/2$ 以确保渲染过程中使用足够的像素。

光栅化。 我们遵循了 3DGS [Kerbl等人 2023] 中类似的光栅化过程。首先，为每个高斯基元计算屏幕空间边界框。然后，根据其中心的深度对二维高斯进行排序，并根据其边界框将它们组织成瓦片。最后，使用体积 alpha 混合从前到后集成 alpha 加权外观：

$$\mathbf{c}(\mathbf{x}) = \sum_{i=1} \mathbf{c}_i \, \alpha_i \, \hat{\mathcal{G}}_i(\mathbf{u}(\mathbf{x})) \prod_{j=1}^{i-1} (1 - \alpha_j \, \hat{\mathcal{G}}_j(\mathbf{u}(\mathbf{x}))) \tag{12}$$

当累积的不透明度达到饱和时，迭代过程终止。

5 训练

我们的二维高斯方法在几何建模中虽然有效，但仅通过光度损失优化时会导致噪声较大的重建结果，这是3D重建任务中固有的挑战 [Barron等人 2022b; Yu等人 2022b; Zhang等人 2020]。为了缓解这一问题并改进几何重建，我们引入了两个正则化项：深度畸变和法向一致性。

深度畸变。 与NeRF不同，3DGS的体积渲染不考虑相交高斯原素之间的距离。因此，高斯扩散可能导致相似的颜色和深度渲染。这与表面渲染不同，表面渲染中光线与第一个可见表面恰好相交一次。为了缓解这一问题，我们借鉴了 Mip-NeRF360 [Barron等人 2022a]，并提出了一种深度畸变损失来通过最小化光线喷溅交点之间的距离来集中沿光线的权重分布：

$$\mathcal{L}_d = \sum_{i,j} \omega_i \omega_j |z_i - z_j| \tag{13}$$

其中 $\omega_i = \alpha_i \hat{}\mathrm{G}_i(\mathbf{u}(\mathbf{x})) \prod_{j=1}^{i-1}(1 - \alpha_j \hat{}\mathrm{G}j(\mathbf{u}(\mathbf{x})))$ 是第 i–个交点的混合权重，而 z_i 是交点的深度。与 Mip-NeRF360 中的畸变损失不同，其中 z_i 是采样点之间的距离且不被优化，我们的方法通过调整交点深度 z_i 直接鼓励 splats 的集中。请注意，我们使用 CUDA 以类似于 [Sun 等人 2022b] 的方式高效地实现了此正则化项。

法向一致性。 由于我们的表示基于 2D 高斯表面元素，我们必须确保所有 2D splats 都与实际表面局部对齐。在体渲染的上下文中，当沿光线可能存在多个半透明 surfels 时，我们考虑实际表面位于交点 $\mathbf{p}_s$ 的中点处，其中累积的不透明度达到 0.5。然后我们按照以下方式将 splats 的法向与深度图的梯度对齐：

$$\mathcal{L}_n = \sum_i \omega_i (1 - \mathbf{n}_i^{\mathrm{T}} \mathbf{N}) \tag{14}$$

在沿光线的相交splat上遍历 i 索引处， ω 表示交点处的混合权重，$\mathbf{n}_i$ 表示朝向相机的splat法线，而$\mathbf{N}$ 是通过深度图梯度估计的法线。具体来说，$\mathbf{N}$ 通过以下方式从附近的深度点计算有限差分得出：

$$\mathbf{N}(x, y) = \frac{\nabla_x \mathbf{p}_s \times \nabla_y \mathbf{p}_s}{|\nabla_x \mathbf{p}_s \times \nabla_y \mathbf{p}_s|} \tag{15}$$

通过将splat法线与估计的表面法线对齐，我们确保2D splats在局部近似实际物体表面。

最终损失。 最后，我们使用一组姿态图像从初始稀疏点云中优化我们的模型。我们最小化以下损失函数：

$$\mathcal{L} = \mathcal{L}_c + \alpha \mathcal{L}_d + \beta \mathcal{L}_n \tag{16}$$

其中Lc 是一个结合L$_1$ 和来自 [Kerbl等人 2023], 的D-SSIM项的RGB重建损失，而L$_d$ 和Ln 是正则化项。我们为有界场景设置 $\alpha = 1000$ ，为无界场景设置 $\alpha = 100$ ，为所有场景设置 $\beta = 0.05$。

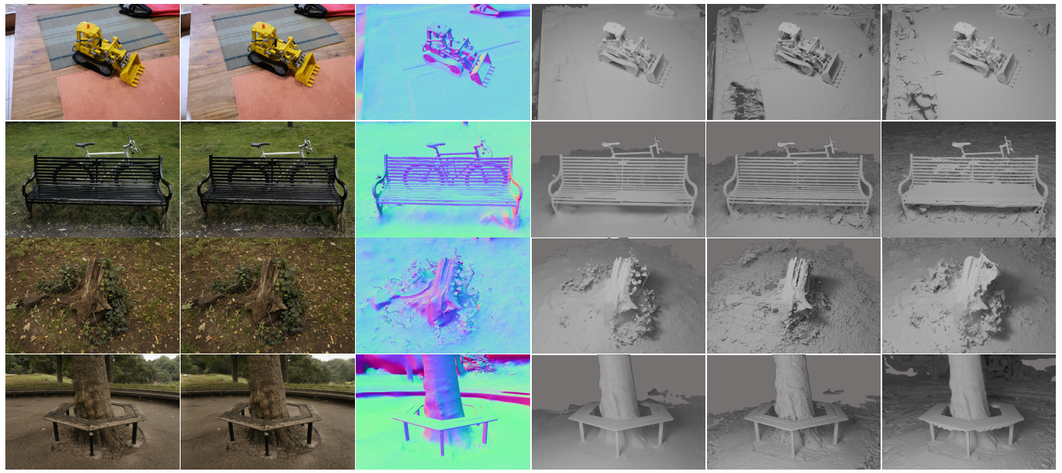
6 实验部分

我们现在展示我们2D高斯splat重建方法的评估，包括与先前最先进的隐式和显式方法的显式和几何比较。然后，我们分析所提出组件的贡献。

6.1 实现方案

我们使用自定义的CUDA内核实现了我们的2D高斯基元，基于3DGS [Kerbl等人 2023]的框架。我们扩展了渲染器以输出用于正则化的深度失真图、深度图和法线图（详见补充材料附录A和B的详细计算）。在训练过程中，我们根据3DGS中的自适应控制策略增加2D高斯基元的数量。由于我们的方法不直接依赖于投影2D中心的梯度，因此我们将3D中心 $\mathbf{p}_k$ 的梯度投影到屏幕空间作为近似值。类似地，我们采用梯度阈值0.0002，并在每3000步中移除不透明度低于0.05的splat。我们在单个GTX RTX3090 GPU上进行了所有实验。

网格提取。 要从重建的2D splats中提取网格，我们使用splat的深度值投影到像素上渲染训练视图的深度图，并利用截断符号距离融合（TSDF）融合重建深度图，使用Open3D [Zhou等人 2018]的方法。我们在TSDF融合期间将体素大小设置为0.004，截断阈值设置为0.02。我们还扩展了



Ground truth | Ours (color) | Ours (normal) | Ours | 3DGS | SuGaR

Fig. 4. Visual comparisons (test-set view) between our method, 3DGS [Kerbl et al. 2023], and SuGaR [Guédon and Lepetit 2023] using scenes from an real-world dataset [Barron et al. 2022b]. Our method excels at synthesizing geometrically accurate radiance fields and surface reconstruction, outperforming 3DGS and SuGaR in capturing sharp edges and intricate details.

Table 1. Quantitative comparison on the DTU Dataset [Jensen et al. 2014]. Our 2DGS achieves the highest reconstruction accuracy among other methods and provides 100× speed up compared to the SDF based baselines.

		24	37	40	55	63	65	69	83	97	105	106	110	114	118	122	Mean	Time
implicit	NeRF [Mildenhall et al. 2021]	1.90	1.60	1.85	0.58	2.28	1.27	1.47	1.67	2.05	1.07	0.88	2.53	1.06	1.15	0.96	1.49	> 12h
	VolSDF [Yariv et al. 2021]	1.14	1.26	0.81	0.49	1.25	0.70	0.72	1.29	1.18	0.70	0.66	1.08	0.42	0.61	0.55	0.86	>12h
	NeuS [Wang et al. 2021]	1.00	1.37	0.93	0.43	1.10	0.65	0.57	1.48	1.09	0.83	0.52	1.20	0.35	0.49	0.54	0.84	>12h
explicit	3DGS [Kerbl et al. 2023]	2.14	1.53	2.08	1.68	3.49	2.21	1.43	2.07	2.22	1.75	1.79	2.55	1.53	1.52	1.50	1.96	11.2 m
	SuGaR [Guédon and Lepetit 2023]	1.47	1.33	1.13	0.61	2.25	1.71	1.15	1.63	1.62	1.07	0.79	2.45	0.98	0.88	0.79	1.33	∼ 1h
	2DGS-15k (Ours)	0.48	0.92	0.42	0.40	1.04	0.83	0.83	1.36	1.27	0.76	0.72	1.63	0.40	0.76	0.60	0.83	5.5 m
	2DGS-30k (Ours)	0.48	0.91	0.39	0.39	1.01	0.83	0.81	1.36	1.27	0.76	0.70	1.40	0.40	0.76	0.52	0.80	10.9 m

Table 2. Quantitative results on the Tanks and Temples Dataset [Knapitsch et al. 2017]. We report the F1 score and training time.

	NeuS	Geo-Neus	Neurlangelo	SuGaR	3DGS	Ours
Barn	0.29	0.33	0.70	0.14	0.13	0.41
Caterpillar	0.29	0.26	0.36	0.16	0.08	0.23
Courthouse	0.17	0.12	0.28	0.08	0.09	0.16
Ignatius	0.83	0.72	0.89	0.33	0.04	0.51
Meetingroom	0.24	0.20	0.32	0.15	0.01	0.17
Truck	0.45	0.45	0.48	0.26	0.19	0.45
Mean	0.38	0.35	0.50	0.19	0.09	0.32
Time	>24h	>24h	>24h	>1h	14.3 m	15.5 m

original 3DGS to render depth and employ the same technique for surface reconstruction for a fair comparison.

6.2 Comparison

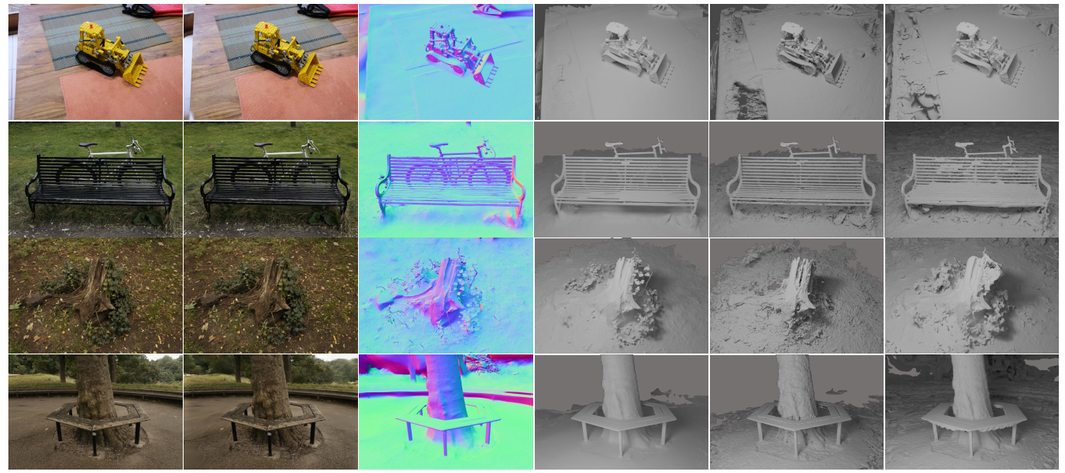
Dataset. We evaluate the performance of our method on various datasets, including DTU [Jensen et al. 2014], Tanks and Temples [Knapitsch et al. 2017], and Mip-NeRF360 [Barron et al. 2022a]. The DTU dataset comprises 15 scenes, each with 49 or 69 images of

Table 3. Performance comparison between 2DGS (ours), 3DGS and SuGaR on the DTU dataset [Jensen et al. 2014]. We report the averaged chamfer distance, PSNR (training-set view), reconstruction time, and model size.

	CD ↓	PSNR ↑	Time ↓	MB (Storage) ↓
3DGS [Kerbl et al. 2023]	1.96	**35.76**	11.2 m	113
SuGaR [Guédon and Lepetit 2023]	1.33	34.57	∼1 h	1247
2DGS-15k (Ours)	0.83	33.42	**5.5 m**	**52**
2DGS-30k (Ours)	**0.80**	34.52	10.9 m	**52**

resolution 1600 × 1200. We use Colmap [Schönberger and Frahm 2016] to generate a sparse point cloud for each scene and downsample the images into a resolution of 800 × 600 for efficiency. We use the same training process for 3DGS [Kerbl et al. 2023] and SuGaR [Guédon and Lepetit 2023] for a comparison.

Geometry Reconstruction. In Table 1 and Table 3, we compare our geometry reconstruction to SOTA implicit (i.e., NeRF [Mildenhall et al. 2020], VolSDF [Yariv et al. 2021], and NeuS [Wang et al. 2021]), explicit (i.e., 3DGS [Kerbl et al. 2023] and concurrent work SuGaR [Guédon and Lepetit 2023]) methods on Chamfer distance



真值 | 我们的（彩色） | 我们的（法线） | Ours | 3DGS | SuGaR

图4. 视觉比较（测试集视图）显示了我们、3DGS [Kerbl等人 2023], 以及SuGaR [Guédon和Lepetit 2023]使用来自真实世界数据集 [Barron等人 2022b] 的场景。我们的方法在合成几何精确的辐射场和表面重建方面表现出色，在捕捉锐利边缘和复杂细节方面优于3DGS和SuGaR。

表 1. 定量比较 DTU数据集 [Jensen等人 2014]. 我们的2DGS在重建精度上优于其他方法，提供 100× 比较于 基于SDF的基线。

		24	37	40	55	63	65	69	83	97	105	106	110	114	118	122	Mean	Time
implicit	NeRF [Mildenhall 等人. 2021]	1.90	1.60	1.85	0.58	2.28	1.27	1.47	1.67	2.05	1.07	0.88	2.53	1.06	1.15	0.96	1.49	> 12h
	VolSDF [Yariv等人 2021]	1.14	1.26	0.81	0.49	1.25	0.70	0.72	1.29	1.18	0.70	0.66	1.08	0.42	0.61	0.55	0.86	>12h
	NeuS [Wang等人 2021]	1.00	1.37	0.93	0.43	1.10	0.65	0.57	1.48	1.09	0.83	0.52	1.20	0.35	0.49	0.54	0.84	>12h
explicit	3DGS [Kerbl等人 2023]	2.14	1.53	2.08	1.68	3.49	2.21	1.43	2.07	2.22	1.75	1.79	2.55	1.53	1.52	1.50	1.96	11.2 m
	SuGaR [Guédon和Lepetit 2023]	1.47	1.33	1.13	0.61	2.25	1.71	1.15	1.63	1.62	1.07	0.79	2.45	0.98	0.88	0.79	1.33	∼ 1h
	2DGS-15k (我们的)	0.48	0.92	0.42	0.40	1.04	0.83	0.83	1.36	1.27	0.76	0.72	1.63	0.40	0.76	0.60	0.83	5.5 m
	2DGS-30k (我们的)	0.48	0.91	0.39	0.39	1.01	0.83	0.81	1.36	1.27	0.76	0.70	1.40	0.40	0.76	0.52	0.80	10.9 m

表2。在坦克与寺庙数据集上的定量结果 [Knapitsch等人 2017]。我们报告了F1分数和训练时间。

	NeuS	Geo-Neus	Neurlangelo	SuGaR	3DGS	Ours
Barn	0.29	0.33	0.70	0.14	0.13	0.41
毛毛虫	0.29	0.26	0.36	0.16	0.08	0.23
法院	0.17	0.12	0.28	0.08	0.09	0.16
伊格纳修斯	0.83	0.72	0.89	0.33	0.04	0.51
会议室	0.24	0.20	0.32	0.15	0.01	0.17
卡车	0.45	0.45	0.48	0.26	0.19	0.45
Mean	0.38	0.35	0.50	0.19	0.09	0.32
Time	>24h	>24h	>24h	>1h	14.3 m	15.5 m

原始3DGS以渲染深度，并采用相同的技术进行表面重建，以进行公平的比较。

6.2 比较

数据集。 我们在多个数据集上评估了我们的方法性能，包括DTU [Jensen等人 2014], 坦克与神庙 [Knapitsch等人 2017],以及Mip-NeRF 360 [Barron等人2022a]。DTU数据集包含15个场景，每个场景有49或69张图像，

表3。在DTU数据集 [上，2DGS（我们的）、3DGS和SuGaR的性能比较 2014]。我们报告了平均Chamfer距离、PSNR（训练集视图）、重建时间和模型大小。

	CD ↓	PSNR ↑	时间 ↓	MB (存储) ↓
3DGS [Kerbl等人 2023]	1.96	**35.76**	11.2 m	113
SuGaR [Guédon和Lepetit 2023]	1.33	34.57	∼1 h	1247
2DGS-15k (我们的)	0.83	33.42	**5.5米**	**52**
2DGS-30k（我们的）	**0.80**	34.52	10.9米	**52**

分辨率 1600 × 1200。我们使用Colmap [Schönberger和Frahm 2016] 为每个场景生成稀疏点云，并将图像下采样到 800 × 600 分辨率以提高效率。我们使用与3DGS [Kerbl等人 2023] 以及SuGaR [Guédon和Lepetit 2023] 相同的训练过程进行比较。

几何重建。 在表1和表3中，我们将我们的几何重建与SOTA隐式（即NeRF [Milden-hall等人 2020], VolSDF [Yariv等人 2021], 和NeuS [Wang等人2021]）、显式（即3DGS [Kerbl等人 2023] 和并行工作 SuGaR [Guédon和Lepetit 2023]）方法在Chamfer距离上进行了比较

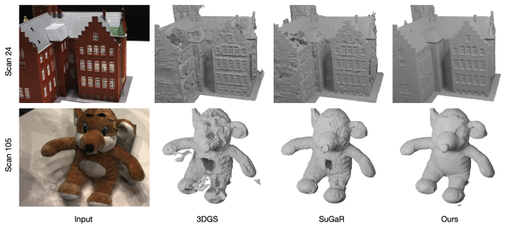


Fig. 5. Qualitative comparison on the DTU benchmark [Jensen et al. 2014]. Our 2DGS produces detailed and noise-free surfaces.

Table 4. Quantitative results on Mip-NeRF 360 [Barron et al. 2022a] dataset. All scores of the baseline methods are directly taken from their papers whenever available. We report the performance of 3DGS, SuGaR and ours using 30k iterations.

	Outdoor Scene			Indoor scene		
	PSNR ↑	SSIM ↑	LIPPS ↓	PSNR ↑	SSIM ↑	LIPPS ↓
NeRF	21.46	0.458	0.515	26.84	0.790	0.370
Deep Blending	21.54	0.524	0.364	26.40	0.844	0.261
Instant NGP	22.90	0.566	0.371	29.15	0.880	0.216
MERF	23.19	0.616	0.343	27.80	0.855	0.271
BakedSDF	22.47	0.585	0.349	27.06	0.836	0.258
MipNeRF360	24.47	0.691	0.283	31.72	0.917	0.180
Mobile-NeRF	21.95	0.470	0.470	-	-	-
SuGaR	22.93	0.629	0.356	29.43	0.906	0.225
3DGS	24.64	0.731	0.234	30.41	0.920	0.189
2DGS (Ours)	24.34	0.717	0.246	30.40	0.916	0.195

and training time using the DTU dataset. Our method outperforms all compared methods in terms of Chamfer distance. Moreover, as shown in Table 2, 2DGS achieves competitive results with SDF models (i.e., NeuS [Wang et al. 2021] and Geo-Neus [Fu et al. 2022]) on the TnT dataset, and significantly better reconstruction than explicit reconstruction methods (i.e., 3DGS and SuGaR). Notably, our model demonstrates exceptional efficiency, offering a reconstruction speed that is approximately 100 times faster compared to implicit reconstruction methods and more than 3 times faster than the concurrent work SuGaR. Our approach can also achieve qualitatively better reconstructions with more appearance and geometry details and fewer outliers, as shown in Figure 5. Moreover, SDF-based reconstruction methods require predefining the spherical size for initialization, which plays a critical role in the success of SDF reconstruction. By contrast, our method leverages radiance field based geometry modeling and is less sensitive to initialization. We include the full geometry reconstruction results for both DTU and TnT in Figure 9 and Figure 10.

Appearance Reconstruction. Our method represents 3D scenes as radiance fields, providing high-quality novel view synthesis. In this section, we compare our novel view renderings using the Mip-NeRF360 dataset against baseline approaches, as shown in Table 4 and Figure 4. Note that, since the ground truth geometry is not available in the Mip-NeRF360 dataset and we hence focus on quantitative comparison. Remarkably, our method consistently achieves competitive NVS results across state-of-the-art techniques while providing geometrically accurate surface reconstruction. We include the appearance rendering results in Figure 11.

Table 5. Quantitative studies for the regularization terms and mesh extraction methods on the DTU dataset.

	Accuracy ↓	Completion ↓	Average ↓
A. w/o normal consistency	1.35	1.13	1.24
B. w/o depth distortion	0.89	0.87	0.88
C. w / expected depth	0.88	1.01	0.94
D. w / SPSR	1.25	0.89	1.07
E. Full Model	0.79	0.86	0.83

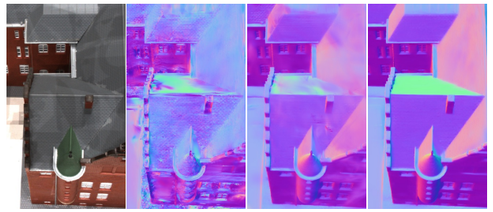


Fig. 6. Qualitative studies for the regularization effects. From left to right – input image, surface normals without normal consistency, without depth distortion, and our full model. Disabling the normal consistency loss leads to noisy surface orientations; conversely, omitting depth distortion regularization results in blurred surface normals. The complete model, employing both regularizations, successfully captures sharp and flat features.

6.3 Ablations

In this section, we isolate the design choices and measure their effect on reconstruction quality, including regularization terms and mesh extraction. We conduct experiments on the DTU dataset [Jensen et al. 2014] with 15k iterations and report the reconstruction accuracy, completeness and average reconstruction quality. The quantitative effect of the choices is reported in Table 5. Additional baseline comparisons can be found in Appendix C of the supplemental material.

Regularization. We first examine the effects of the proposed normal consistency and depth distortion regularization terms. Our model (Table 5 E) provides the best performance when applying both regularization terms. We observe that disabling the normal consistency (Table 5 A) can lead to incorrect orientation, as shown in Figure 6 A. Additionally, the absence of depth distortion (Table 5 B) results in a noisy surface, as shown in Figure 6 B.

Mesh Extraction. We now analyze our choice for mesh extraction. Our full model (Table 5 E) utilizes TSDF fusion for mesh extraction with median depth. One alternative option is to use the expected depth instead of the median depth. However, it yields worse reconstructions as it is more sensitive to outliers, as shown in Table 5 C. Further, our approach surpasses screened Poisson surface reconstruction (SPSR)(Table 5 D) [Kazhdan and Hoppe 2013] using 2D Gaussians' center and normal as inputs, due to SPSR's inability to incorporate the opacity and the size of 2D Gaussian primitives.

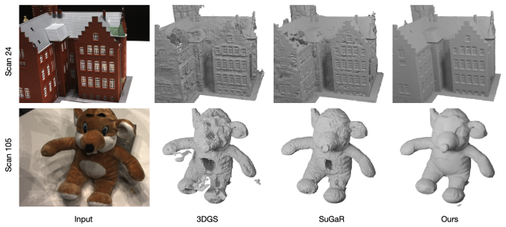


图5. 在DTU基准上的定性比较 [Jensen等人 2014]. 我们的2DGS生成了细节丰富且无噪声的表面。

表4. 在Mip-NeRF 360 [Barron等人2022a]数据集上的定量结果. 所有基线方法的分数在可用时直接取自其论文. 我们报告了3DGS、SuGaR和我们的性能，使用 30k 迭代。

	户外场景			室内场景		
	PSNR ↑	SSIM ↑	LIPPS ↓	PSNR ↑	SSIM ↑	LIPPS ↓
NeRF	21.46	0.458	0.515	26.84	0.790	0.370
深度融合	21.54	0.524	0.364	26.40	0.844	0.261
即时 NGP	22.90	0.566	0.371	29.15	0.880	0.216
MERF	23.19	0.616	0.343	27.80	0.855	0.271
烘焙 SDF	22.47	0.585	0.349	27.06	0.836	0.258
MipNeRF360	24.47	0.691	0.283	31.72	0.917	0.180
移动端 NeRF	21.95	0.470	0.470	-	-	-
SuGaR	22.93	0.629	0.356	29.43	0.906	0.225
3DGS	24.64	0.731	0.234	30.41	0.920	0.189
2DGS (我们的)	24.34	0.717	0.246	30.40	0.916	0.195

以及使用DTU数据集的训练时间。我们的方法在Chamfer距离方面优于所有比较方法。此外，如表2所示，2DGS在TnT数据集上与SDF模型（即NeuS [Wang等人 2021] 和 Geo-Neus [Fu等人 2022]）取得了具有竞争力的结果，并且比显式重建方法（即3DGS和SuGaR）的重建效果显著更好。值得注意的是，我们的模型表现出卓越的效率，其重建速度比隐式重建方法快约100倍，比并行工作SuGaR快3倍以上。我们的方法还可以实现定性的更好重建，具有更多外观和几何细节，以及更少的异常值，如图5所示。此外，基于SDF的重建方法需要在初始化时预定义球形大小，这对SDF重建的成功起着至关重要的作用。相比之下，我们的方法利用基于辐射场的几何建模，对初始化不太敏感。我们在图9和图10中包含了DTU和TnT的完整几何重建结果。

外观重建。 我们的方法将3D场景表示为辐射场，提供高质量的新视角合成。在本节中，我们使用Mip-NeRF360数据集比较我们新视角渲染的基线方法，如表4和图4所示。请注意，由于真值几何形状不是

可在Mip-NeRF360数据集中获得，因此我们着重于定量比较。值得注意的是，我们的方法始终如一地实现了

表5. 正则化项和网格提取方法在DTU数据集上的定量研究。

	精度 ↓	完整性 ↓	平均 ↓
A. 无法向一致性	1.35	1.13	1.24
B. 无深度失真	0.89	0.87	0.88
C. w / 期望深度	0.88	1.01	0.94
D. w / SPSR	1.25	0.89	1.07
E. 完整模型	0.79	0.86	0.83

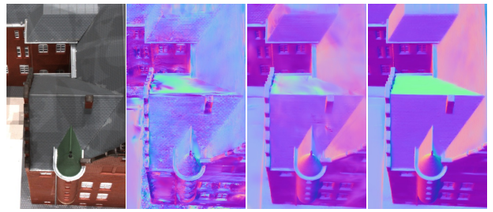


图6. 定性研究正则化效果。从左到右——输入图像、无法向一致性的表面法线、无深度失真、以及我们的完整模型。禁用法向一致性损失会导致噪声的表面方向；反之，省略深度失真正则化会导致模糊的表面法线。完整模型采用两种正则化，成功捕捉了锐利和平坦特征。

在当前最先进的技术中取得具有竞争力的 NVS 结果，同时提供几何上精确的表面重建。我们在图 11 中包含了外观渲染的结果。

6.3 消融实验

在本节中，我们隔离设计选择并测量其对重建质量的影响，包括正则化项和网格提取。我们在DTU数据集 [Jensenet al. 2014] 上使用 15k 次迭代进行实验，并报告重建精度、完整性和平均重建质量。这些选择的影响量化结果在表5中报告。附加基线比较可在补充材料的附录C中找到。

正则化。 我们首先考察了所提出的法向一致性和深度失真正则化项的影响。我们的模型（表5 E）在应用这两个正则化项时性能最佳。我们观察到禁用法向一致性（表5 A）会导致方向错误，如图6 A所示。此外，缺少深度失真（表5 B）会导致表面出现噪声，如图6 B所示。

网格提取。 我们现在分析我们选择网格提取的方法。我们的完整模型（表5 E）使用基于中值深度的TSDF融合进行网格提取。另一种替代选项是使用期望深度而不是中值深度。然而，它会产生更差的重建效果，因为它对离群值更敏感，如表5 C所示。此外，我们的方法超越了筛选泊松表面重建（SPSR）（表5 D）[Kazhdan和Hoppe 2013] 使用2D高斯基元的中心点和法线作为输入，这是因为SPSR无法结合2D高斯基元的透明度和大小。

7 CONCLUSION

We presented 2D Gaussian splatting, a novel approach for geometrically accurate radiance field reconstruction. We utilized 2D Gaussian primitives for 3D scene representation, facilitating accurate and view consistent geometry modeling and rendering. We proposed two regularization techniques to further enhance the reconstructed geometry. Extensive experiments on several challenging datasets verify the effectiveness and efficiency of our method.

Limitations. While our method successfully delivers accurate appearance and geometry reconstruction for a wide range of objects and scenes, we also discuss its limitations: First, we assume surfaces with full opacity and extract meshes from multi-view depth maps. This can pose challenges in accurately handling semi-transparent surfaces, such as glass, due to their complex light transmission properties, as shown in Figure 12. Secondly, our current densification strategy favors texture-rich over geometry-rich areas, occasionally leading to less accurate representations of fine geometric structures. A more effective densification strategy could mitigate this issue. Finally, our regularization often involves a trade-off between image quality and geometry, and can potentially lead to over-smoothing in certain regions.

ACKNOWLEDGMENTS

BH and SG are supported by NSFC #62172279, #61932020, Program of Shanghai Academic Research Leader. ZY, AC and AG are supported by the ERC Starting Grant LEGO-3D (850533) and DFG EXC number 2064/1 - project number 390727645.

REFERENCES

Kara-Ali Aliev, Artem Sevastopolsky, Maria Kolos, Dmitry Ulyanov, and Victor Lempitsky. 2020. Neural point-based graphics. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXII 16*. Springer, 696–712.

Jonathan T. Barron, Ben Mildenhall, Matthew Tancik, Peter Hedman, Ricardo Martin-Brualla, and Pratul P. Srinivasan. 2021. Mip-NeRF: A Multiscale Representation for Anti-Aliasing Neural Radiance Fields. *ICCV* (2021).

Jonathan T Barron, Ben Mildenhall, Dor Verbin, Pratul P Srinivasan, and Peter Hedman. 2022a. Mip-nerf 360: Unbounded anti-aliased neural radiance fields. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 5470–5479.

Jonathan T. Barron, Ben Mildenhall, Dor Verbin, Pratul P. Srinivasan, and Peter Hedman. 2022b. Mip-NeRF 360: Unbounded Anti-Aliased Neural Radiance Fields. *CVPR* (2022).

Jonathan T. Barron, Ben Mildenhall, Dor Verbin, Pratul P. Srinivasan, and Peter Hedman. 2023. Zip-NeRF: Anti-Aliased Grid-Based Neural Radiance Fields. *ICCV* (2023).

James F Blinn. 1977. A homogeneous formulation for lines in 3 space. In *Proceedings of the 4th annual conference on Computer graphics and interactive techniques*. 237–241.

Mario Botsch, Alexander Hornung, Matthias Zwicker, and Leif Kobbelt. 2005. High-quality surface splatting on today's GPUs. In *Proceedings Eurographics/IEEE VGTC Symposium Point-Based Graphics, 2005.* IEEE, 17–141.

Anpei Chen, Zexiang Xu, Andreas Geiger, Jingyi Yu, and Hao Su. 2022. TensoRF: Tensorial Radiance Fields. In *European Conference on Computer Vision (ECCV)*.

Hanlin Chen, Chen Li, and Gim Hee Lee. 2023b. NeuSG: Neural Implicit Surface Reconstruction with 3D Gaussian Splatting Guidance. *arXiv preprint arXiv:2312.00846* (2023).

Zhiqin Chen, Thomas Funkhouser, Peter Hedman, and Andrea Tagliasacchi. 2023a. Mobilenerf: Exploiting the polygon rasterization pipeline for efficient neural field rendering on mobile architectures. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 16569–16578.

Zhang Chen, Zhong Li, Liangchen Song, Lele Chen, Jingyi Yu, Junsong Yuan, and Yi Xu. 2023c. NeuRBF: A Neural Fields Representation with Adaptive Radial Basis Functions. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 4182–4194.

Sara Fridovich-Keil, Alex Yu, Matthew Tancik, Qinhong Chen, Benjamin Recht, and Angjoo Kanazawa. 2022. Plenoxels: Radiance Fields without Neural Networks. In *CVPR*.

Qiancheng Fu, Qingshan Xu, Yew-Soon Ong, and Wenbing Tao. 2022. Geo-Neus: Geometry-Consistent Neural Implicit Surfaces Learning for Multi-view Reconstruction. *Advances in Neural Information Processing Systems (NeurIPS)* (2022).

Jian Gao, Chun Gu, Youtian Lin, Hao Zhu, Xun Cao, Li Zhang, and Yao Yao. 2023. Relightable 3D Gaussian: Real-time Point Cloud Relighting with BRDF Decomposition and Ray Tracing. *arXiv:2311.16043* (2023).

Antoine Guédon and Vincent Lepetit. 2023. SuGaR: Surface-Aligned Gaussian Splatting for Efficient 3D Mesh Reconstruction and High-Quality Mesh Rendering. *arXiv preprint arXiv:2311.12775* (2023).

Peter Hedman, Pratul P Srinivasan, Ben Mildenhall, Jonathan T Barron, and Paul Debevec. 2021. Baking neural radiance fields for real-time view synthesis. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 5875–5884.

Wenbo Hu, Yuling Wang, Lin Ma, Bangbang Yang, Lin Gao, Xiao Liu, and Yuewen Ma. 2023. Tri-MipRF: Tri-Mip Representation for Efficient Anti-Aliasing Neural Radiance Fields. In *ICCV*.

Eldar Insafutdinov and Alexey Dosovitskiy. 2018. Unsupervised learning of shape and pose with differentiable point clouds. *Advances in neural information processing systems* 31 (2018).

Rasmus Jensen, Anders Dahl, George Vogiatzis, Engin Tola, and Henrik Aanæs. 2014. Large scale multi-view stereopsis evaluation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 406–413.

Yingwenqi Jiang, Jiadong Tu, Yuan Liu, Xifeng Gao, Xiaoxiao Long, Wenping Wang, and Yuexin Ma. 2023. GaussianShader: 3D Gaussian Splatting with Shading Functions for Reflective Surfaces. *arXiv preprint arXiv:2311.17977* (2023).

Michael Kazhdan and Hugues Hoppe. 2013. Screened poisson surface reconstruction. *ACM Transactions on Graphics (ToG)* 32, 3 (2013), 1–13.

Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 2023. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Transactions on Graphics* 42, 4 (July 2023). https://repo-sam.inria.fr/fungraph/3d-gaussian-splatting/

Leonid Keselman and Martial Hebert. 2022. Approximate differentiable rendering with algebraic surfaces. In *European Conference on Computer Vision*. Springer, 596–614.

Leonid Keselman and Martial Hebert. 2023. Flexible techniques for differentiable rendering with 3d gaussians. *arXiv preprint arXiv:2308.14737* (2023).

Arno Knapitsch, Jaesik Park, Qian-Yi Zhou, and Vladlen Koltun. 2017. Tanks and Temples: Benchmarking Large-Scale Scene Reconstruction. *ACM Transactions on Graphics* 36, 4 (2017).

Georgios Kopanas, Julien Philip, Thomas Leimkühler, and George Drettakis. 2021. Point-Based Neural Rendering with Per-View Optimization. In *Computer Graphics Forum*, Vol. 40. Wiley Online Library, 29–43.

Christoph Lassner and Michael Zollhofer. 2021. Pulsar: Efficient sphere-based neural rendering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 1440–1449.

Zhaoshuo Li, Thomas Müller, Alex Evans, Russell H Taylor, Mathias Unberath, Ming-Yu Liu, and Chen-Hsuan Lin. 2023. Neuralangelo: High-Fidelity Neural Surface Reconstruction. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

Zhihao Liang, Qi Zhang, Ying Feng, Ying Shan, and Kui Jia. 2023. GS-IR: 3D Gaussian Splatting for Inverse Rendering. *arXiv preprint arXiv:2311.16473* (2023).

Lingjie Liu, Jiatao Gu, Kyaw Zaw Lin, Tat-Seng Chua, and Christian Theobalt. 2020. Neural Sparse Voxel Fields. *NeurIPS* (2020).

Jonathon Luiten, Georgios Kopanas, Bastian Leibe, and Deva Ramanan. 2024. Dynamic 3D Gaussians: Tracking by Persistent Dynamic View Synthesis. In *3DV*.

Lars Mescheder, Michael Oechsle, Michael Niemeyer, Sebastian Nowozin, and Andreas Geiger. 2019. Occupancy Networks: Learning 3D Reconstruction in Function Space. In *Conference on Computer Vision and Pattern Recognition (CVPR)*.

Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. 2020. NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis. In *ECCV*.

Ben Mildenhall, Pratul P Srinivasan, Matthew Tancik, Jonathan T Barron, Ravi Ramamoorthi, and Ren Ng. 2021. Nerf: Representing scenes as neural radiance fields for view synthesis. *Commun. ACM* 65, 1 (2021), 99–106.

Thomas Müller, Alex Evans, Christoph Schied, and Alexander Keller. 2022. Instant Neural Graphics Primitives with a Multiresolution Hash Encoding. *ACM Trans. Graph.* 41, 4, Article 102 (July 2022), 15 pages.

Michael Niemeyer, Lars Mescheder, Michael Oechsle, and Andreas Geiger. 2020. Differentiable Volumetric Rendering: Learning Implicit 3D Representations without 3D Supervision. In *Conference on Computer Vision and Pattern Recognition (CVPR)*.

Michael Oechsle, Songyou Peng, and Andreas Geiger. 2021. UNISURF: Unifying Neural Implicit Surfaces and Radiance Fields for Multi-View Reconstruction. In *International Conference on Computer Vision (ICCV)*.

Jeong Joon Park, Peter Florence, Julian Straub, Richard Newcombe, and Steven Lovegrove. 2019. DeepSDF: Learning Continuous Signed Distance Functions for Shape Representation. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

Hanspeter Pfister, Matthias Zwicker, Jeroen Van Baar, and Markus Gross. 2000. Surfels: Surface elements as rendering primitives. In *Proceedings of the 27th annual conference*

7 结论

我们提出了2D高斯喷溅，这是一种用于几何精确辐射场重建的新方法。我们使用2D高斯基元进行3D场景表示，从而实现精确且视图一致的几何建模和渲染。我们提出了两种正则化技术，以进一步增强重建的几何效果。在几个具有挑战性的数据集上的大量实验验证了我们方法的有效性和效率。

局限性。虽然我们的方法成功地实现了对各种物体和场景的外观和几何重建，但我们也讨论了它的局限性：首先，我们假设表面是完全不透明的，并从多视图深度图中提取网格。这可能会在处理半透明表面（如玻璃）时遇到挑战，因为它们具有复杂的透光特性，如图12所示。其次，我们当前的稠密化策略更倾向于纹理丰富的区域而不是几何丰富的区域，偶尔会导致对精细几何结构的表示不够准确。更有效的稠密化策略可以缓解这个问题。最后，我们的正则化通常涉及图像质量和几何形状之间的权衡，并且可能会在某些区域导致过度平滑。

致谢

BH和SG由国家自然科学基金（#62172279）、（#61932020）和上海市学术研究带头人计划资助。ZY、AC和AG由ERC启动资助计划LEGO-3D（850533）和DFG卓越集群计划编号2064/1 - 项目编号390727645资助。

参考文献

卡拉-阿里·阿利耶夫，阿特姆·谢瓦斯托波尔斯基，玛丽亚·科洛斯，德米特里·乌利亚诺夫，和维克多·莱姆皮茨基。2020。基于神经的点图形学。在<样式 id='1'>计算机视觉–ECCV 2020：第16届欧洲会议，格拉斯哥，英国，2020年8月23日至28日，论文集，第XXII部分 16</样式>。Springer，696–712。乔纳森·T·巴隆，本·米德尔哈尔，马修·T·坦西克，彼得·黑德曼，里卡多·马丁-布鲁亚拉，和普拉图尔·P·斯里尼瓦桑。2021。Mip-NeRF：用于抗锯齿神经辐射场的多尺度表示。<样式 id='3'>ICCV </样式>(2021)。乔纳森·T·巴隆，本·米德尔哈尔，多·韦尔宾，普拉图尔·P·斯里尼瓦桑，和彼得·黑德曼。2022a。Mip-nerf 360：无界抗锯齿神经辐射场。在<样式 id='5'>IEEE/CVF计算机视觉与模式识别会议论文集</样式>。5470–5479。乔纳森·T·巴隆，本·米德尔哈尔，多·韦尔宾，普拉图尔·P·斯里尼瓦桑，和彼得·黑德曼。2022b。Mip-NeRF 360：无界抗锯齿神经辐射场。<样式 id='7'>CVPR </样式>(2022)。乔纳森·T·巴隆，本·米德尔哈尔，多·韦尔宾，普拉图尔·P·斯里尼瓦桑，和彼得·黑德曼。2023。Zip-NeRF：抗锯齿基于网格的神经辐射场。<样式 id='9'>ICCV </样式>(2023)。詹姆斯·F·布林。1977。三维空间中线的齐次公式。在<样式 id='11'>第4届计算机图形学和交互技术年会的论文集</样式>。237–241。马里奥·博什，亚历山大·霍恩乌ن，马蒂亚斯·齐维克尔，和莱夫·科布尔特。2005。当今GPU上的高质量表面喷溅。在<样式 id='13'>2005年欧洲图形学/IEEE VGTC点图形学研讨会论文集</样式>IEEE，17–141。陈安培，徐泽祥，安德烈亚斯·盖格，余静怡，和苏浩。2022。TensoRF：张量辐射场。在<样式 id='15'>欧洲计算机视觉会议 (ECCV)</样式>。陈汉林，李晨，和吉姆·希·李。2023b。NeuSG：具有3D高斯喷溅指导的神经隐式表面重建。<样式 id='17'>arXiv预印本 arXiv:2312.00846 </样式>(2023)。陈志钦，托马斯·芬克豪泽，彼得·黑德曼，和安德烈亚·塔利亚萨奇。2023a。Mobilenerf：利用多边形光栅化管道在移动架构上高效渲染神经场。在<样式 id='19'>IEEE/CVF计算机视觉与模式识别会议论文集</样式>。16569–16578。陈张，李钟，宋良辰，陈磊磊，余静怡，袁俊松，和徐毅。2023c。NeuRBF：具有自适应径向基函数的神经场表示。在<样式 id='21'>IEEE/CVF国际计算机视觉会议论文集</样式>。4182–4194。萨拉·弗里多维奇-凯尔，亚历克斯·余，马修·坦西克，陈庆红，本杰明·瑞奇，和安乔·卡纳扎瓦。2022。plenoxels：无需神经网络的辐射场。在<样式 id='23'>CVPR</样式>。

Qiancheng Fu, Qingshan Xu, Yew-Soon Ong, and Wenbing Tao. 2022. Geo-Neus: 几何一致性神经隐式曲面学习用于多视图重建 (Geo-Neus: Geometry-Consistent Neural Implicit Surfaces Learning for Multi-view Reconstruction). 神经信息处理系统进展 (*NeurIPS*) (2022). Jian Gao, Chun Gu, Youtian Lin, Hao Zhu, Xun Cao, Li Zhang, and Yao Yao. 2023. 可重新光照的3D高斯: 基于BRDF分解和光线追踪的实时点云重新光照 (Re-lightable 3D Gaussian: Real-time Point Cloud Relighting with BRDF Decomposition and Ray Tracing). *arXiv:2311.16043* (2023). Antoine Guédon and Vincent Lepetit. 2023. SuGaR: 表面对齐的3D高斯喷溅用于高效的3D网格重建和高质量网格渲染 (SuGaR: Surface-Aligned Gaussian Splatting for Efficient 3D Mesh Reconstruction and High-Quality Mesh Rendering). *arXiv preprint arXiv:2311.12775* (2023). Peter Hedman, Pratul P Srinivasan, Ben Mildenhall, Jonathan T Barron, and Paul De- bevec. 2021. 烘焙神经辐射场以实现实时视图合成. 在 ***IEEE/CVF***国际计算机视觉会议论文集. 5875–5884. Wenbo Hu, Yuling Wang, Lin Ma, Bangbang Yang, Lin Gao, Xiao Liu, and Yuewen Ma. 2023. Tri-MipRF: 三重Mip表示用于高效抗走样神经辐射场 (Tri-MipRF: Tri-Mip Representation for Efficient Anti-Aliasing Neural Radiance Fields). 在 *ICCV*. Eldar Insafutdinov and Alexey Dosovitskiy. 2018. 无监督学习形状和姿态使用可微分的点云. 神经信息处理系统进展 31 (2018). Rasmus Jensen, Anders Dahl, George Vogiatzis, Engin Tola, and Henrik Aanæs. 2014. 大规模多视图立体评估. 在 ***IEEE***计算机视觉与模式识别会议论文集. 406–413. Yingwenqi Jiang, Jiadong Tu, Yuan Liu, Xifeng Gao, Xiaoxiao Long, Wenping Wang, and Yuexin Ma. 2023. GaussianShader: 用于反射表面的带阴影函数的3D高斯喷溅 (GaussianShader: 3D Gaussian Splatting with Shading Functions for Reflective Surfaces). *arXiv preprint arXiv:2311.17977* (2023). Michael Kazhdan and Hugues Hoppe. 2013. 筛选泊松表面重建. ***ACM***图形学汇刊 ***(ToG)*** 32, 3 (2013), 1–13. Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 2023. 3D高斯喷溅用于实时辐射场渲染. ***ACM***图形学汇刊 42, 4 (2023年7月). https://repo-sam.inria.fr/fungraph/3d-gaussian-splatting/ Leonid Keselman and Martial Hebert. 2022. 基于代数曲面的近似可微分渲染. 在 欧洲计算机视觉会议. Springer, 596–614. Leonid Keselman and Martial Hebert. 2023. 用于可微分渲染的3D高斯的灵活技术. *arXiv preprint arXiv:2308.14737* (2023). Arno Knapitsch, Jaesik Park, Qian-Yi Zhou, and Vladlen Koltun. 2017. 坦克和神庙: 大规模场景重建基准测试 (Tanks and Temples: Benchmarking Large-Scale Scene Reconstruction). ***ACM***图形学汇刊 36, 4 (2017). Georgios Kopanas, Julien Philip, Thomas Leimkühler, and George Drettakis. 2021. 基于点的神经渲染与视图优化 (Point- Based Neural Rendering with Per-View Optimization). 在 计算机图形学论坛, Vol. 40. Wiley Online Library, 29–43. Christoph Lassner and Michael Zollhofer. 2021. Pulsar: 基于球面的高效神经渲染 (Pulsar: Efficient sphere-based neural rendering). 在 ***IEEE/CVF***国际计算机视觉与模式识别会议论文集. 1440–1449. Zhaoshuo Li, Thomas Müller, Alex Evans, Russell H Taylor, Mathias Unberath, Ming- Yu Liu, and Chen-Hsuan Lin. 2023. Neuralangelo: 高保真神经表面重建 (Neuralangelo: High-Fidelity Neural Surface Reconstruction). 在 ***IEEE***国际计算机视觉与模式识别会议 ***(CVPR)***. Zhihao Liang, Qi Zhang, Ying Feng, Ying Shan, and Kui Jia. 2023. GS-IR: 用于逆向渲染的3D高斯喷溅 (GS-IR: 3D Gaussian Splatting for Inverse Rendering). *arXiv preprint arXiv:2311.16473* (2023). Lingjie Liu, Jiatao Gu, Kyaw Zaw Lin, Tat-Seng Chua, and Christian Theobalt. 2020. 神经稀疏体素场 (Neural Sparse Voxel Fields). *NeurIPS* (2020). Jonathon Luiten, Georgios Kopanas, Bastian Leibe, and Deva Ramanan. 2024. 动态3D高斯: 通过持久动态视图合成进行跟踪 (Dynamic 3D Gaussians: Tracking by Persistent Dynamic View Synthesis). 在 *3DV*. Lars Mescheder, Michael Oechsle, Michael Niemeyer, Sebastian Nowozin, and Andreas Geiger. 2019. 占用网络: 在函数空间中学习3D重建 (Occupancy Networks: Learning 3D Reconstruction in Function Space). 在 计算机视觉与模式识别会议 ***(CVPR)***. Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ra- mamoorthi, and Ren Ng. 2020. NeRF: 将场景表示为神经辐射场以实现视图合成 (NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis). 在 *ECCV*. Ben Mildenhall, Pratul P Srinivasan, Matthew Tancik, Jonathan T Barron, Ravi Ra- mamoorthi, and Ren Ng. 2021. Nerf: 将场景表示为神经辐射场以实现视图合成. ***ACM*** 通讯65, 1 (2021), 99–106. Thomas Müller, Alex Evans, Christoph Schied, and Alexander Keller. 2022. 带多分辨率哈希编码的即时神经图形基元 (Instant Neural Graphics Primitives with a Multiresolution Hash Encoding). *ACM Trans. Graph.* 41, 4, Article 102 (2022年7月), 15页. Michael Niemeyer, Lars Mescheder, Michael Oechsle, and Andreas Geiger. 2020. 可微分体积渲染: 无需3D监督学习隐式3D表示 (Differentiable Volumetric Rendering: Learning Implicit 3D Representations without 3D Supervision). 在 计算机视觉与模式识别会议 ***(CVPR)***. Michael Oechsle, Songyou Peng, and Andreas Geiger. 2021. UNISURF: 统一神经隐式曲面和辐射场用于多视图重建 (UNISURF: Unifying Neural Implicit Surfaces and Radiance Fields for Multi-View Reconstruction). 在 国际计算机视觉会议 ***(ICCV)***. Jeong Joon Park, Peter Florence, Julian Straub, Richard Newcombe, and Steven Love- grove. 2019. DeepSDF: 学习用于形状表示的连续符号距离函数 (DeepSDF: Learning Continuous Signed Distance Functions for Shape Representation). 在 ***IEEE***计算机视觉与模式识别会议 ***(CVPR)***. Hanspeter Pfister, Matthias Zwicker, Jeroen Van Baar, and Markus Gross. 2000. Surfels: 作为渲染基元的表面元素 (Surfels: Surface elements as rendering primitives). 在 第27届年度会议论文集

on Computer graphics and interactive techniques. 335–342.
Shenhan Qian, Tobias Kirschstein, Liam Schoneveld, Davide Davoli, Simon Giebenhain, and Matthias Nießner. 2023. GaussianAvatars: Photorealistic Head Avatars with Rigged 3D Gaussians. *arXiv preprint arXiv:2312.02069* (2023).
Christian Reiser, Songyou Peng, Yiyi Liao, and Andreas Geiger. 2021. KiloNeRF: Speeding up Neural Radiance Fields with Thousands of Tiny MLPs. In *International Conference on Computer Vision (ICCV)*.
Christian Reiser, Rick Szeliski, Dor Verbin, Pratul Srinivasan, Ben Mildenhall, Andreas Geiger, Jon Barron, and Peter Hedman. 2023. Merf: Memory-efficient radiance fields for real-time view synthesis in unbounded scenes. *ACM Transactions on Graphics (TOG)* 42, 4 (2023), 1–12.
Darius Rückert, Linus Franke, and Marc Stamminger. 2022. Adop: Approximate differentiable one-pixel point rendering. *ACM Transactions on Graphics (ToG)* 41, 4 (2022), 1–14.
Johannes Lutz Schönberger and Jan-Michael Frahm. 2016. Structure-from-Motion Revisited. In *Conference on Computer Vision and Pattern Recognition (CVPR)*.
Johannes Lutz Schönberger, Enliang Zheng, Marc Pollefeys, and Jan-Michael Frahm. 2016. Pixelwise View Selection for Unstructured Multi-View Stereo. In *European Conference on Computer Vision (ECCV)*.
Thomas Schöps, Torsten Sattler, and Marc Pollefeys. 2019. Surfelmeshing: Online surfel-based mesh reconstruction. *IEEE transactions on pattern analysis and machine intelligence* 42, 10 (2019), 2494–2507.
Yahao Shi, Yanmin Wu, Chenming Wu, Xing Liu, Chen Zhao, Haocheng Feng, Jingtuo Liu, Liangjun Zhang, Jian Zhang, Bin Zhou, Errui Ding, and Jingdong Wang. 2023. GIR: 3D Gaussian Inverse Rendering for Relightable Scene Factorization. *Arxiv* (2023). arXiv:2312.05133
Christian Sigg, Tim Weyrich, Mario Botsch, and Markus H Gross. 2006. GPU-based ray-casting of quadratic surfaces.. In *PBG@ SIGGRAPH*. 59–65.
Cheng Sun, Min Sun, and Hwann-Tzong Chen. 2022a. Direct Voxel Grid Optimization: Super-fast Convergence for Radiance Fields Reconstruction. In *CVPR*.
Cheng Sun, Min Sun, and Hwann-Tzong Chen. 2022b. Improved Direct Voxel Grid Optimization for Radiance Fields Reconstruction. *arxiv cs.GR 2206.05085* (2022).
Peng Wang, Lingjie Liu, Yuan Liu, Christian Theobalt, Taku Komura, and Wenping Wang. 2021. NeuS: Learning Neural Implicit Surfaces by Volume Rendering for Multi-view Reconstruction. *Advances in Neural Information Processing Systems* 34 (2021), 27171–27183.
Yiming Wang, Qin Han, Marc Habermann, Kostas Daniilidis, Christian Theobalt, and Lingjie Liu. 2023. NeuS2: Fast Learning of Neural Implicit Surfaces for Multi-view Reconstruction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*.
Tim Weyrich, Simon Heinzle, Timo Aila, Daniel B Fasnacht, Stephan Oetiker, Mario Botsch, Cyril Flaig, Simon Mall, Kaspar Rohrer, Norbert Felber, et al. 2007. A hardware architecture for surface splatting. *ACM Transactions on Graphics (TOG)* 26, 3 (2007), 90–es.
Thomas Whelan, Renato F Salas-Moreno, Ben Glocker, Andrew J Davison, and Stefan Leutenegger. 2016. ElasticFusion: Real-time dense SLAM and light source estimation. *The International Journal of Robotics Research* 35, 14 (2016), 1697–1716.
Olivia Wiles, Georgia Gkioxari, Richard Szeliski, and Justin Johnson. 2020. SynSin: End-to-end View Synthesis from a Single Image. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
Tianyi Xie, Zeshun Zong, Yuxing Qiu, Xuan Li, Yutao Feng, Yin Yang, and Chenfanfu Jiang. 2023. PhysGaussian: Physics-Integrated 3D Gaussians for Generative Dynamics. *arXiv preprint arXiv:2311.12198* (2023).
Yunzhi Yan, Haotong Lin, Chenxu Zhou, Weijie Wang, Haiyang Sun, Kun Zhan, Xianpeng Lang, Xiaowei Zhou, and Sida Peng. 2023. Street Gaussians for Modeling Dynamic Urban Scenes. (2023).
Yao Yao, Zixin Luo, Shiwei Li, Tian Fang, and Long Quan. 2018. MVSNet: Depth Inference for Unstructured Multi-view Stereo. *European Conference on Computer Vision (ECCV)* (2018).
Lior Yariv, Jiatao Gu, Yoni Kasten, and Yaron Lipman. 2021. Volume rendering of neural implicit surfaces. *Advances in Neural Information Processing Systems* 34 (2021), 4805–4815.
Lior Yariv, Peter Hedman, Christian Reiser, Dor Verbin, Pratul P. Srinivasan, Richard Szeliski, Jonathan T. Barron, and Ben Mildenhall. 2023. BakedSDF: Meshing Neural SDFs for Real-Time View Synthesis. *arXiv* (2023).
Lior Yariv, Yoni Kasten, Dror Moran, Meirav Galun, Matan Atzmon, Basri Ronen, and Yaron Lipman. 2020. Multiview Neural Surface Reconstruction by Disentangling Geometry and Appearance. *Advances in Neural Information Processing Systems* 33 (2020).
Wang Yifan, Felice Serena, Shihao Wu, Cengiz Öztireli, and Olga Sorkine-Hornung. 2019. Differentiable surface splatting for point-based geometry processing. *ACM Transactions on Graphics (TOG)* 38, 6 (2019), 1–14.
Alex Yu, Ruilong Li, Matthew Tancik, Hao Li, Ren Ng, and Angjoo Kanazawa. 2021. PlenOctrees for Real-time Rendering of Neural Radiance Fields. In *ICCV*.
Zehao Yu, Anpei Chen, Bozidar Antic, Songyou Peng, Apratim Bhattacharyya, Michael Niemeyer, Siyu Tang, Torsten Sattler, and Andreas Geiger. 2022a. SDFStudio: A Unified Framework for Surface Reconstruction. https://github.com/autonomousvision/sdfstudio
Zehao Yu, Anpei Chen, Binbin Huang, Torsten Sattler, and Andreas Geiger. 2024. Mip-Splatting: Alias-free 3D Gaussian Splatting. *Conference on Computer Vision and Pattern Recognition (CVPR)* (2024).
Zehao Yu and Shenghua Gao. 2020. Fast-MVSNet: Sparse-to-Dense Multi-View Stereo With Learned Propagation and Gauss-Newton Refinement. In *Conference on Computer Vision and Pattern Recognition (CVPR)*.
Zehao Yu, Songyou Peng, Michael Niemeyer, Torsten Sattler, and Andreas Geiger. 2022b. MonoSDF: Exploring Monocular Geometric Cues for Neural Implicit Surface Reconstruction. *Advances in Neural Information Processing Systems (NeurIPS)* (2022).
Kai Zhang, Gernot Riegler, Noah Snavely, and Vladlen Koltun. 2020. NeRF++: Analyzing and Improving Neural Radiance Fields. *arXiv:2010.07492* (2020).
Qian-Yi Zhou, Jaesik Park, and Vladlen Koltun. 2018. Open3D: A Modern Library for 3D Data Processing. *arXiv:1801.09847* (2018).
Wojciech Zielonka, Timur Bagautdinov, Shunsuke Saito, Michael Zollhöfer, Justus Thies, and Javier Romero. 2023. Drivable 3D Gaussian Avatars. (2023). arXiv:2311.08581 [cs.CV]
Matthias Zwicker, Hanspeter Pfister, Jeroen Van Baar, and Markus Gross. 2001a. EWA volume splatting. In *Proceedings Visualization, 2001. VIS'01.* IEEE, 29–538.
Matthias Zwicker, Hanspeter Pfister, Jeroen Van Baar, and Markus Gross. 2001b. Surface splatting. In *Proceedings of the 28th annual conference on Computer graphics and interactive techniques*. 371–378.
Matthias Zwicker, Jussi Rasanen, Mario Botsch, Carsten Dachsbacher, and Mark Pauly. 2004. Perspective accurate splatting. In *Proceedings-Graphics Interface*. 247–254.

A DETAILS OF DEPTH DISTORTION

While Barron et al. [Barron et al. 2022b] calculates the distortion loss with samples on the ray, we operate Gaussian primitives, where the intersected depth may not be ordered. To this end, we adopt an $\mathcal{L}_2$ loss and transform the intersected depth z to NDC space to down-weight distant Gaussian primitives, $m = \mathrm{NDC}(z)$, with near and far plane empirically set to 0.2 and 1000. We implemented our depth distortion loss based on [Sun et al. 2022b], also powered by tile-based rendering. Here we show that the nested algorithm can be implemented in a single forward pass:

$$\begin{aligned}\mathcal{L} &= \sum_{i=0}^{N-1}\sum_{j=0}^{i-1}\omega_i\omega_j(m_i-m_j)^2 \\ &= \sum_{i=0}^{N-1}\omega_i\left(m_i^2\sum_{j=0}^{i-1}\omega_j + \sum_{j=0}^{i-1}\omega_j m_j^2 - 2m_i\sum_{j=0}^{i-1}\omega_j m_j\right) \\ &= \sum_{i=0}^{N-1}\omega_i\left(m_i^2 A_{i-1} + D_{i-1}^2 - 2m_i D_{i-1}\right),\end{aligned} \tag{17}$$

where $A_i = \sum_{j=0}^{i}\omega_j$, $D_i = \sum_{j=0}^{i}\omega_j m_j$ and $D_i^2 = \sum_{j=0}^{i}\omega_j m_j^2$.

Specifically, we let $e_i = \left(m_i^2 A_{i-1} + D_{i-1}^2 - 2m_i D_{i-1}\right)$ so that the distortion loss can be "rendered" as $\mathcal{L}_i = \sum_{j=0}^{i}\omega_j e_j$. Here, $\mathcal{L}_i$ measures the depth distortion up to the i-th Gaussian. During marching Gaussian front-to-back, we simultaneously accumulate A_i, D_i and D_i^2, preparing for the next distortion computation $\mathcal{L}_{i+1}$. Similarly, the gradient of the depth distortion can be back-propagated to the primitives back-to-front. Different from implicit methods where m are the pre-defined sampled depth and non-differentiable, we additionally back-propagate the gradient through the intersection m, encouraging the Gaussians to move tightly together directly.

在计算机图形学和交互技术领域。 335–342. 钱申涵，科希斯坦因·托比亚斯，肖内费尔德·利亚姆，达沃利·达维德，吉本海因·西蒙，和Nießner·马蒂亚斯。 2023年。GaussianAvatars：具有可配置 3D 高斯的逼真头像。 ***arXiv*** 预印本 ***arXiv:2312.02069*** (2023年)。Christian Reiser，彭松友，廖一艺，和安德烈亚斯·盖格。 2021年。 KiloNeRF：使用数千个微型MLP加速神经辐射场。 在 国际计算机视觉会议（***ICCV***）。 Christian Reiser，Szeliski·里克，Verbin·多，Srinivasan·普拉图尔，Mildenhall·本，安德烈亚斯·盖格，Barron·乔恩，和Hedman·彼得。 2023年。 Merf：用于无界场景中实时视图合成的内存高效辐射场。 *ACM* 图形学交易（ ***TOG*** ） 42，4 (2023 年)，1–12。 Darius Rückert，Franke· 林努斯，和Stamminger·马克。 2022年。 Adop：近似可微分的单像素点渲染。 *ACM*图形学交易（***ToG***） 41，4 (2022年)，1–14。 Schönberger·约翰内斯·路德维希，和Frahm·简·迈克尔。 2016年。 运动结构重访。 在 计算机视觉与模式识别会议（***CVPR***）。 Schönberger·约翰内斯·路德维希，Zheng·恩良，Pollefeys·马克，和Frahm·简·迈克尔。 2016年。 像素级视图选择用于非结构化多视图立体。 在 欧洲计算机视觉会议（***ECCV***）。 Schöps·托马斯，Sattler·托尔sten，和Pollefeys·马克。 2019 年。 Surfelmeshing：基于表面单元的在线网格重建。 ***IEEE*** 模式分析和机器智能交易 42，10 (2019年)，2494–2507。 Shi·亚豪，Wu·严敏，Wu·陈明，Liu·行，Zhao·陈，Feng·郝成，Liu·景拓，Zhang·梁军，Zhang·简，Zhou·斌，Ding·尔瑞，和Wang·景东。 2023年。GIR：3D高斯逆渲染用于可重光照场景分解。 *Arxiv* (2023年)。 arXiv:2312.05133 Christian Sigg，Weyrich·蒂姆，Botsch·马里奥，和Gross·马克斯·H。 2006年。 基于GPU的二次曲面射线投射。 在 *PBG@ SIGGRAPH*。 59–65。 Sun·程，Sun·敏，和Chen·Hwann-Tzong。 2022a。 直接体素网格优化：辐射场重建的超快速收敛。 在 *CVPR*。 Sun·程，Sun·敏，和Chen·Hwann-Tzong。 2022b。 用于辐射场重建的改进直接体素网格优化。 *arxiv cs.GR2206.05085* (2022年)。 Wang·彭，Liu·凌杰，Liu·袁，Theobalt·Christian，Komura·Taku，和王·文平。 2021年。 NeuS：通过体积渲染学习神经隐式表面用于多视图重建。 神经信息处理系统进展 34 (2021年)，27171–27183。 Wang·杨明，Han·秦，Habermann·Marc，Daniilidis·Kostas，Theobalt·Christian，和Liu·凌杰。 2023年。 NeuS2：用于多视图重建的快速神经隐式表面学习。 在 ***IEEE/CVF***国际计算机视觉会议论文集（***ICCV***）。 Weyrich·Tim，Heinzle·西蒙，Aila·Timo，Fasnacht·Daniel B，Oetiker·斯蒂芬，Botsch·马里奥，Flaig·西里尔，Mall·西蒙，Rohrer·卡斯帕，Felber·诺伯特，等。 2007年。 用于表面喷溅的硬件架构。 ***ACM***图形学交易（***TOG***） 26，3 (2007年)，90–es。 Whelan·托马斯，Salas-Moreno·Renato F，Glocker·Ben，Davison·Andrew J，和Leutenegger·斯特凡。 2016年。 ElasticFusion：实时密集SLAM和光源估计。 国际机器人研究杂志 35，14 (2016年)，1697–1716。 Wiles·奥利维亚，Gkioxari·乔治亚，Szeliski·Richard，和Johnson·Justin。 2020年。 SynSin：从单图像到端到端视图合成。 在 ***IEEE/CVF***计算机视觉与模式识别会议论文集（***CVPR***）。 Xie·谢天一，Zong·宗泽顺，Qiu·邱宇行，Li·宣，Feng·冯宇涛，Yang·尹阳，和Jiang·蒋晨潘。 2023年。 PhysGaussian：物理集成3D高斯用于生成动态。 ***arXiv***预印本 ***arXiv:2311.12198*** (2023年)。 Yan·云枝，Lin·海彤，Zhou·陈旭，Wang·魏杰，Sun·海阳，Zhan·昆，Lang·西安鹏，Zhou·晓伟，和Peng·四达。 2023年。 Street Gaussians用于建模动态城市场景。 (2023年)。 Yao·姚，Luo·罗子欣，Li·石伟，Fang·田方，和Quan·全。 2018年。 MVSNet：非结构化多视图立体的深度推断。 欧洲计算机视觉会议（***ECCV***） (2018年)。 Yariv·利奥，Gu·Jiatao，Kasten·Yoni，和Lipman·亚伦。 2021年。 神经隐式表面的体积渲染。 神经信息处理系统进展 34 (2021年)，4805–4815。 Yariv·利奥，Hedman·彼得，Reiser·Christian，Verbin·多，Srinivasan·普拉图尔·P，Szeliski·Richard，Barron·Jonathan T，和Mildenhall·本。 2023年。 BakedSDF：用于实时视图合成的神经SDF网格化。 ***Arxiv*** (2023年)。 Yariv·利奥，Kasten·Yoni，Moran·Dror，Galun·Meirav，Atzmon·Matan，Ronen·Basri，和Lipman·亚伦。 2020 年。 通过解耦几何和外观进行多视图神经表面重建。 神经信息处理系统进展 33 (2020年)。 Wang·Yifan，Serena·Felice，Wu·石豪，Öztireli·Cengiz，和Sorkine-Hornung·奥拉。 2019年。 用于点几何处理的可微分表面喷溅。 ***ACM***图形学交易（***TOG***） 38，6 (2019年)，1–14。 Yu·Alex，Li·Ruilong，Tancik·Matthew，Li·郝，Ng·Ren，和Kanazawa·Angjoo。 2021年。 PlenOctrees用于实时渲染神经辐射场。 在 *ICCV*。
余哲豪，陈安培，Bozidar Antic，彭松友，Apratim Bhattacharyya，Michael Niemeyer，唐思宇，Torsten Sattler，以及安德烈亚斯·盖格。2022a。 SDFStudio：一个用于表面重建的统一框架。 https://github.com/autonomousvision/sdfstudio 余哲豪，陈安培，黄斌斌，Torsten Sattler，以及安德烈亚斯·盖格。2024。Mip-Splatting：无伪影3D高斯Splatting。计算机视觉与模式识别会议（***CVPR***）(2024)。余哲豪和高胜华。2020。Fast-MVSNet：稀疏到密集多视图立体学习传播和高斯牛顿细化。在计算机视觉与模式识别会议（***CVPR***）。余哲豪，彭松友，Michael Niemeyer，Torsten Sattler，以及安德烈亚斯·盖格。2022b。MonoSDF：探索用于神经隐式表面重建的单目几何线索。神经信息处理系统进展（***NeurIPS***）(2022)。张凯，Gernot Riegler，Noah Snavely，以及Vladlen Koltun。2020。NeRF++：分析和改进神经辐射场。*arXiv:2010.07492* (2020)。周千一，Jaesik Park，以及Vladlen Koltun。2018。Open3D：现代3D数据处理库。*arXiv:1801.09847* (2018)。Wojciech Zielonka，Timur Bagautdinov，Saito Shunsuke，Michael Zollhöfer，Justus Thies，以及Javier Romero。2023。可驾驶3D高斯头像。(2023)。arXiv:2311.08581 [cs.CV] Matthias Zwicker，Hanspeter Pfister，Jeroen Van Baar，以及Markus Gross。2001a。EWA体积Splatting。在***2001***年可视化会议论文集 ***VIS'01***IEEE，29–538。Matthias Zwicker，Hanspeter Pfister，Jeroen Van Baar，以及Markus Gross。2001b。表面喷溅。在第***28***届计算机图形学和交互技术年会议论文集。371–378。Matthias Zwicker，Jussi Rasanen，Mario Botsch，Carsten Dachsbacher，以及Mark Pauly。2004。透视精确Splatting。在***Graphics Interface***会议论文集。247–254。

深度畸变的详细说明

虽然Barron等人 [Barron等人 2022b] 在光线上计算畸变损失，但我们处理高斯基元，其中相交的深度可能无序。为此，我们采用L_2 损失，并将相交的深度 z 转换为NDC空间以降低远距离高斯基元的权重，$m = \mathrm{NDC}(z)$，近远平面经验设置为0.2和1000。我们基于 [Sun等人 2022b], 实现了深度畸变损失，后者也采用基于瓦片的渲染。这里我们展示了嵌套算法可以在单次前向传递中实现：

$$\begin{aligned}\mathcal{L} &= \sum_{i=0}^{N-1}\sum_{j=0}^{i-1}\omega_i\omega_j(m_i-m_j)^2 \\ &= \sum_{i=0}^{N-1}\omega_i\left(m_i^2\sum_{j=0}^{i-1}\omega_j + \sum_{j=0}^{i-1}\omega_j m_j^2 - 2m_i\sum_{j=0}^{i-1}\omega_j m_j\right) \\ &= \sum_{i=0}^{N-1}\omega_i\left(m_i^2 A_{i-1} + D_{i-1}^2 - 2m_i D_{i-1}\right),\end{aligned} \tag{17}$$

其中 $A_i = \sum_{j=0}^{i}\omega_j$, $D_i = \sum_{j=0}^{i}\omega_j m_j$和 $D_i^2 = \sum_{j=0}^{i}\omega_j m_j^2$。

具体来说，我们让 $e_i = \left(m_i^2 A_{i-1} + D_{i-1}^2 - 2m_i D_{i-1}\right)$，以便失真损失可以被“渲染”为 $\mathrm{L}_i = \sum_{j=0}^{i}\omega_j e_j$。这里，$\mathrm{L}_i$测量高达第 i 个高斯（Gaussian）的深度失真（depth distortion）。在 Marching Gaussian 从前往后推进的过程中，我们同时累积 A_i、D_i 和 D_i^2，为下一次失真计算 L_{i+1} 做准备。类似地，深度失真的梯度也可以从后往前反向传播到基元（primitives）。与隐式方法不同，其中 m 是预定义的采样深度且不可微，我们额外通过交点 m 反向传播梯度，鼓励高斯（Gaussians）紧密地相互靠近。

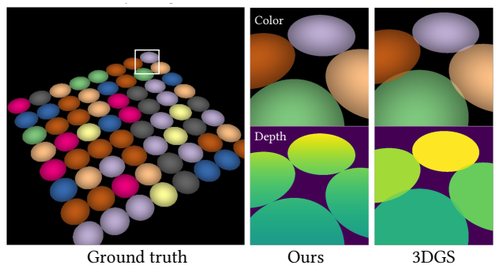


Fig. 7. Visualization of a plane tiled by 2D Gaussians. Affine approximation [Zwicker et al. 2001b] adopted in 3DGS [Kerbl et al. 2023] causes perspective distortion and inaccurate depth, violating normal consistency.

B DEPTH CALCULATIONS

Mean depth: There are two optional depth computations used for our meshing process. The mean (expected) depth is calculated by weighting the intersected depth:

$$z_{\text{mean}} = \sum_i \omega_i z_i / (\sum_i \omega_i + \epsilon) \tag{18}$$

where $\omega_i = T_i \alpha_i \hat{\mathcal{G}}_i(\mathbf{u}(\mathbf{x}))$ is the weight contribution of the i-th Gaussian and $T_i = \prod_{j=1}^{i-1}(1 - \alpha_j \hat{\mathcal{G}}_j(\mathbf{u}(\mathbf{x})))$ measures its visibility. It is important to normalize the depth with the accumulated alpha $A = \sum_i \omega_i$ to ensure that a 2D Gaussian can be rendered as a planar 2D disk in the depth visualization.

Median depth: We compute the median depth as the largest "visible" depth, considering $T_i = 0.5$ as the pivot for surface and free space:

$$z_{\text{median}} = \max\{z_i | T_i > 0.5\}. \tag{19}$$

We find our median depth computation is more robust to [Luiten et al. 2024]. When a ray's accumulated alpha does not reach 0.5, while Luiten et al. sets a default value of 15, our computation selects the last Gaussian, which is more accurate and suitable for training.

C ADDITIONAL BASELINES

In this section, we present additional baselines to ablate the impact of our design choices, as summarized in Table 6. Furthermore, we integrate our meshing approach into the comparison against these baselines for a comprehensive analysis. SuGaR extracts a mesh from depth points utilizing SPSR (Screen Poisson Surface Reconstruction) during the coarse stage, followed by refinement using a mesh renderer. To assess the effect of this meshing strategy, we substituted SPSR with TSDF using the depth maps, followed by an identical refinement stage. However, we found that the depth map generated from their flat Gaussian intersection is sparse and discontinuous. As a result, the adaptation of TSDF with its discontinuous depth map yields inferior results. For 3DGS, we leverage SPSR for mesh generation. Because the 3D Gaussian lacks a surface normal, we treat its normal as a trainable parameter [Gao et al. 2023; Liang et al. 2023] distilled from the depth map, employing the normal consistency regularization. We then utilize all center points for SPSR, resulting in

Table 6. Additional baselines on DTU dataset. All the models are trained with $30k$ iterations.

	Accuracy ↓	Completion ↓	Average ↓
SuGaR	1.48	1.17	1.33
SuGaR + TSDF	2.47	1.90	2.18
3DGS + SPSR (center)	2.05	1.25	1.65
3DGS + TSDF (mean)	1.93	1.99	1.96
2DGS + SPSR (center)	1.25	0.89	1.07
2DGS (affine approx) + TSDF (mean)	0.96	1.20	1.08
2DGS (our rasterizer) + TSDF (mean)	0.79	0.98	0.88
2DGS (our rasterizer) + TSDF (median)	0.78	0.83	0.80

improved overall completion metrics compared to results obtained using TSDF.

Finally, we conduct ablation experiments on our 2DGS. Notably, 2DGS demonstrates enhanced performance by iteratively integrating components such as TSDF, perspective-correct rasterization, and median depth. For the affine approximation baseline, we utilize 3DGS's rasterization method by configuring one scale of the 3D Gaussian to $1e^{-6}$. While affine approximation already yields promising results, integrating the proposed ray-splat intersection scheme results in more accurate depth map generation under perspective projection, as depicted in Figure 7, thus enhancing depth fusion performance.

D ADDITIONAL RESULTS

Our 2D Gaussian Splatting method achieves comparable performance even without the need for regularizations, as Table 7 shows. We have included a detailed breakdown of per-scene metrics for the MipNeRF360 dataset [Barron et al. 2022b] in Table 9. Additionally, we have provided a comparison of our rendered depth maps with those from 3DGS and MipNeRF360 in Figure 8.

Table 7. PSNR scores for Synthetic NeRF dataset. Our model achieve comparable performance without using regularizations.

	Mic	Chair	Ship	Materials	Lego	Drums	Ficus	Hotdog	Mean
Plenoxels	33.26	33.98	29.62	29.14	34.10	25.35	31.83	36.81	31.76
INGP-Base	36.22	35.00	31.10	29.78	36.39	26.02	33.51	37.40	33.18
Mip-NeRF	36.51	35.14	30.41	30.71	35.70	25.48	33.29	37.48	33.09
3DGS	35.36	35.83	30.80	30.00	35.78	26.15	34.87	37.72	33.32
Ours	35.09	35.05	30.60	29.74	35.10	26.05	35.57	37.36	33.07

Table 8. PSNR scores for TnT dataset.

	Barn	Caterpillar	Courthouse	Ignatius	Meetingroom	Truck	Mean
SuGaR	28.63	23.27	23.33	20.72	25.47	24.40	24.16
3DGS	27.99	24.82	23.33	23.95	26.89	25.01	25.33
Ours	28.79	24.23	23.51	23.82	26.15	26.85	25.56

Table 9. PSNR↑, SSIM↑, LIPPS↓ scores for MipNeRF360 dataset.

	bicycle	flowers	garden	stump	treehill	room	counter	kitchen	bonsai	mean
SugaR	23.34	19.54	25.40	25.07	21.30	29.97	27.56	29.41	30.77	25.82
3DGS	25.24	21.52	27.41	26.55	22.49	30.63	28.70	30.32	31.98	27.20
Ours	24.87	21.15	26.95	26.47	22.27	31.06	28.55	30.50	31.52	27.03
SuGaR	0.634	0.499	0.762	0.705	0.546	0.904	0.885	0.902	0.933	0.752
3DGS	0.771	0.605	0.868	0.775	0.638	0.914	0.905	0.922	0.938	0.815
Ours	0.752	0.588	0.852	0.765	0.627	0.912	0.900	0.919	0.933	0.805
SuGaR	0.354	0.407	0.240	0.325	0.452	0.259	0.244	0.178	0.220	0.298
3DGS	0.205	0.336	0.103	0.210	0.317	0.220	0.204	0.129	0.205	0.214
Ours	0.218	0.346	0.115	0.222	0.329	0.223	0.208	0.133	0.214	0.223

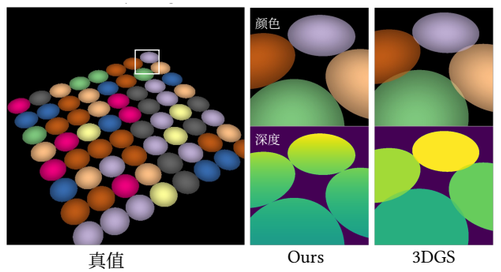


图7. 由2D高斯分布铺成的平面可视化。3DGS [Kerbl等人 2023] 中采用的仿射近似 [Zwicker等人 2001b] 会导致透视失真和不准确的深度，违反法向一致性。

B 深度计算

均值深度：我们的网格化（meshing）过程中有两个可选的深度计算方法。均值（期望）深度通过加权相交深度来计算：

$$z_{\text{mean}} = \sum_i \omega_i z_i / (\sum_i \omega_i + \epsilon) \tag{18}$$

ˆ其中 $\omega_i = T_i \alpha_i \mathrm{G}_i(\mathbf{u}(\mathbf{x}))$ 是第 i 个高斯权重贡献，而 $T_i = \prod_{j=1}^{i-1}(1 - \alpha_j \hat{}\mathrm{G}j(\mathbf{u}(\mathbf{x})))$ 则衡量其可见性。重要的是将深度与累积的 alpha$A = \sum_i \omega_i$ 进行归一化，以确保 2D 高斯可以在深度可视化中渲染为平面 2D 圆盘。

中值深度：我们计算中值深度为最大的“可见”深度，将 $T_i = 0.5$ 作为表面和自由空间的基准：

$$z_{\text{median}} = \max\{z_i | T_i > 0.5\}. \tag{19}$$

我们发现我们的中值深度计算对 [Luiten 等人 2024] 更鲁棒。当一条射线的累积 alpha 未能达到 0.5 时，而 Luiten 等人设置了一个默认值 15，我们的计算会选择最后一个高斯，这更准确且适合训练。

C 额外基准

在本节中，我们提出了额外的基线来消除我们设计选择的影响，如表6所示。此外，我们将我们的网格化方法整合到与这些基线的比较中，以进行全面分析。SuGaR在粗略阶段利用SPSR（屏幕泊松表面重建）从深度点提取网格，然后使用网格渲染器进行细化。为了评估这种网格化策略的影响，我们使用深度图将SPSR替换为TSDF，然后进行相同的细化阶段。然而，我们发现他们平面高斯交集生成的深度图是稀疏且不连续的。因此，使用其不连续的深度图对TSDF进行适配会产生较差的结果。对于3DGS，我们利用SPSR进行网格生成。由于3D高斯缺乏表面法线，我们将其法线视为可训练参数 [Gao等人 2023年；Liang等人 2023]从深度图中提取，采用法向一致性正则化。然后我们使用所有中心点进行SPSR，得到

表6. 在DTU数据集上的额外基线。所有模型均使用 $30k$ 迭代次数进行训练。

	准确率 ↓	完成度 ↓	平均值 ↓
SuGaR	1.48	1.17	1.33
SuGaR + TSDF	2.47	1.90	2.18
3DGS +SPSR (中心)	2.05	1.25	1.65
3DGS +TSDF (均值)	1.93	1.99	1.96
2DGS +SPSR (中心)	1.25	0.89	1.07
2DGS (仿射近似) + TSDF (均值)	0.96	1.20	1.08
2DGS (我们光栅化器) +TSDF (均值)	0.79	0.98	0.88
2DGS (我们光栅化器) +TSDF (中值)	0.78	0.83	0.80

比使用TSDF获得的结果更好的整体完成指标。

最后，我们对我们的2DGS进行消融实验。值得注意的是，2DGS通过迭代集成TSDF、透视校正光栅化和中值深度等组件，表现出增强的性能。对于仿射近似基线，我们通过将3D高斯的一个尺度配置为 $1e^{-6}$来使用3DGS的光栅化方法。虽然仿射近似已经产生了令人鼓舞的结果，但集成所提出的射线喷溅交点方案在透视投影下生成了更精确的深度图，如图7所示，从而增强了深度融合性能。

D 其他结果

我们的2D高斯splatting方法即使不需要正则化也能达到相当的性能，如表7所示。我们已在表9中包含了针对MipNeRF360数据集 [Barron等人 2022b] 的详细场景指标。此外，我们还提供了我们渲染的深度图与3DGS和MipNeRF360的深度图在图8中的对比。

表7. 合成NeRF数据集的PSNR分数。我们的模型在不使用正则化的情况下也能达到相当的性能。

	Mic	Chair	Ship	Materials	Lego	Drums	Ficus	Hotdog	均值	
plenoxels	33.26	33.98	29.62	29.14	34.10	25.35	31.83	36.81	31.76	
INGP-Base	36.22	35.00	31.10	29.78	36.39	26.02	33.51	37.40	33.18	
Mip-NeRF	36.51	35.14	30.41	30.71	35.70	25.48	33.29	37.48	33.09	
3DGS	35.36	35.83	30.80	30.00	35.78	26.15	34.87	37.72	33.32	
Ours	35.09	35.05	30.60	29.74	35.10	26.05	35.57	37.36	33.07	

表8. TnT数据集的PSNR分数。

	Barn	毛毛虫	法院	伊格纳修斯		会议室	卡车	Mean
SuGaR	28.63	23.27		23.33	20.72	25.47	24.40	24.16
3DGS	27.99	24.82		23.33	23.95	26.89	25.01	25.33
Ours	28.79	24.23		23.51	23.82	26.15	26.85	25.56

表9. MipNeRF360数据集的PSNR↑, SSIM↑, LIPPS↓ 分数。

	自行车	花园	花园	拐杖	树山	房间	计数	厨房	篮子	平均	
SuGaR	23.34	19.54	25.40	25.07	21.30	29.97	27.56	29.41	30.77	25.82	
3DGS	25.24	21.52	27.41	26.55	22.49	30.63	28.70	30.32	31.98	27.20	
Ours	24.87	21.15	26.95	26.47	22.27	31.06	28.55	30.50	31.52	27.03	
SuGaR	0.634	0.499	0.762	0.705	0.546	0.904	0.885	0.902	0.933	0.752	
3DGS	0.771	0.605	0.868	0.775	0.638	0.914	0.905	0.922	0.938	0.815	
Ours	0.752	0.588	0.852	0.765	0.627	0.912	0.900	0.919	0.933	0.805	
SuGaR	0.354	0.407	0.240	0.325	0.452	0.259	0.244	0.178	0.220	0.298	
3DGS	0.205	0.336	0.103	0.210	0.317	0.220	0.204	0.129	0.205	0.214	
Ours	0.218	0.346	0.115	0.222	0.329	0.223	0.208	0.133	0.214	0.223	

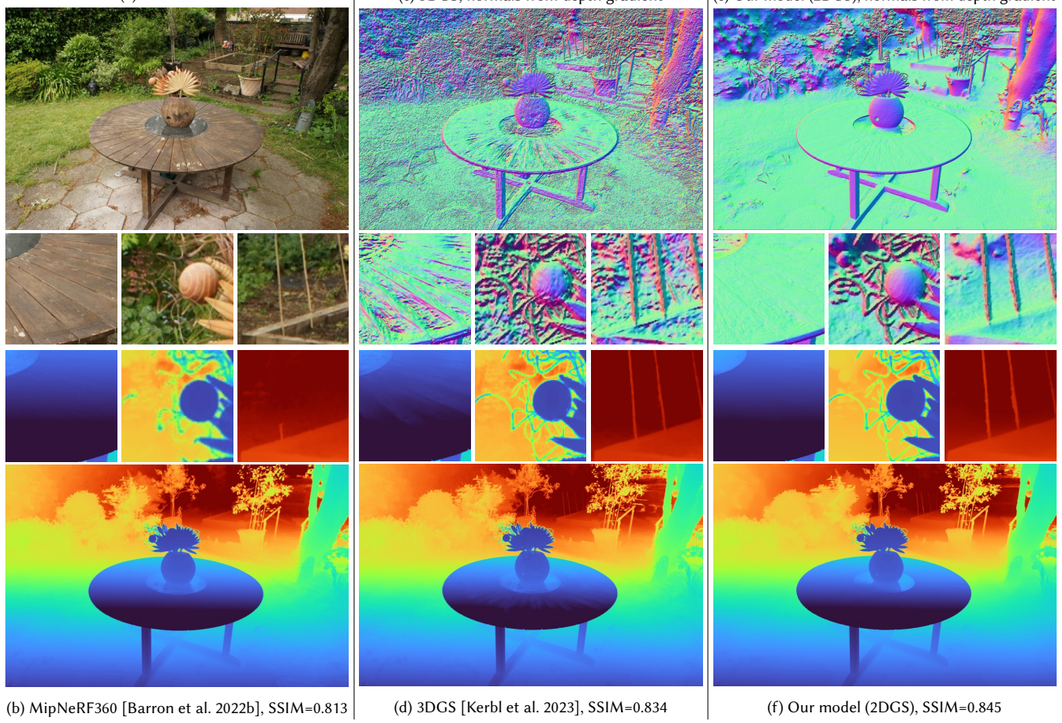


Fig. 8. We visualize the depth maps generated by MipNeRF360 [Barron et al. 2022b], 3DGS [Kerbl et al. 2023], and our method. The depth maps for 3DGS (d) and 2DGS (f) are rendered using Eq. 18 and visualized following MipNeRF360. To highlight the surface smoothness, we further visualize the normal estimated from depth gradient using Eq. 15 for both 3DGS (c) and ours (e). While MipNeRF360 is capable of producing plausibly smooth depth maps, its sampling process may result in the loss of detailed structures. Both 3DGS and 2DGS excel at modeling thin structures; however, as illustrated in (c) and (e), the depth map of 3DGS exhibits significant noise. In contrast, our approach generates sampled depth points with normals consistent with the rendered normal map (refer to Figure 1b), thereby enhancing depth fusion during the meshing process.

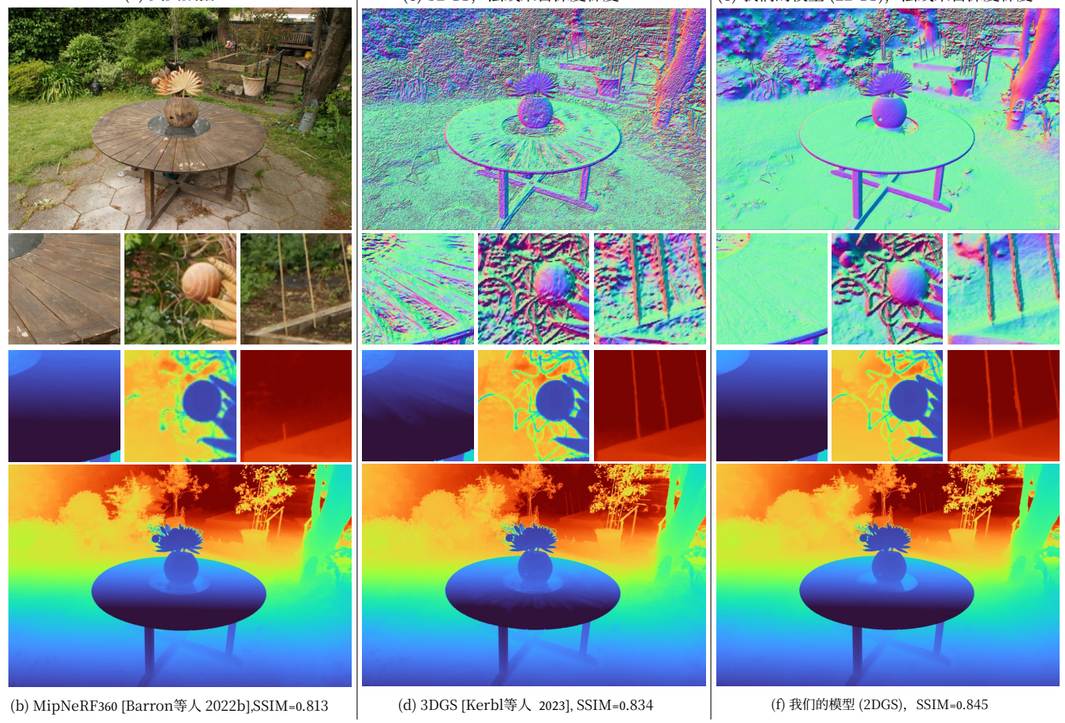


图8。我们可视化了MipNeRF360 [Barron等人 2022b], 32DGS [Kerbl等人 2023], 以及我们方法生成的深度图。3DGS (d) 和 2DGS (f) 的深度图使用公式18渲染，并遵循MipNeRF360进行可视化。为了突出表面平滑性，我们进一步使用公式15对3DGS (c) 和我们方法 (e) 从深度梯度估计法线进行可视化。虽然MipNeRF360能够生成看似平滑的深度图，但其采样过程可能导致细节结构的丢失。3DGS 和 2DGS 都擅长建模细小结构；然而，如(c)和(e)所示，3DGS的深度图存在显著噪声。相比之下，我们方法生成的采样深度点与渲染的法线图（参见图1b）的法线一致，从而在网格化过程中增强了深度融合。

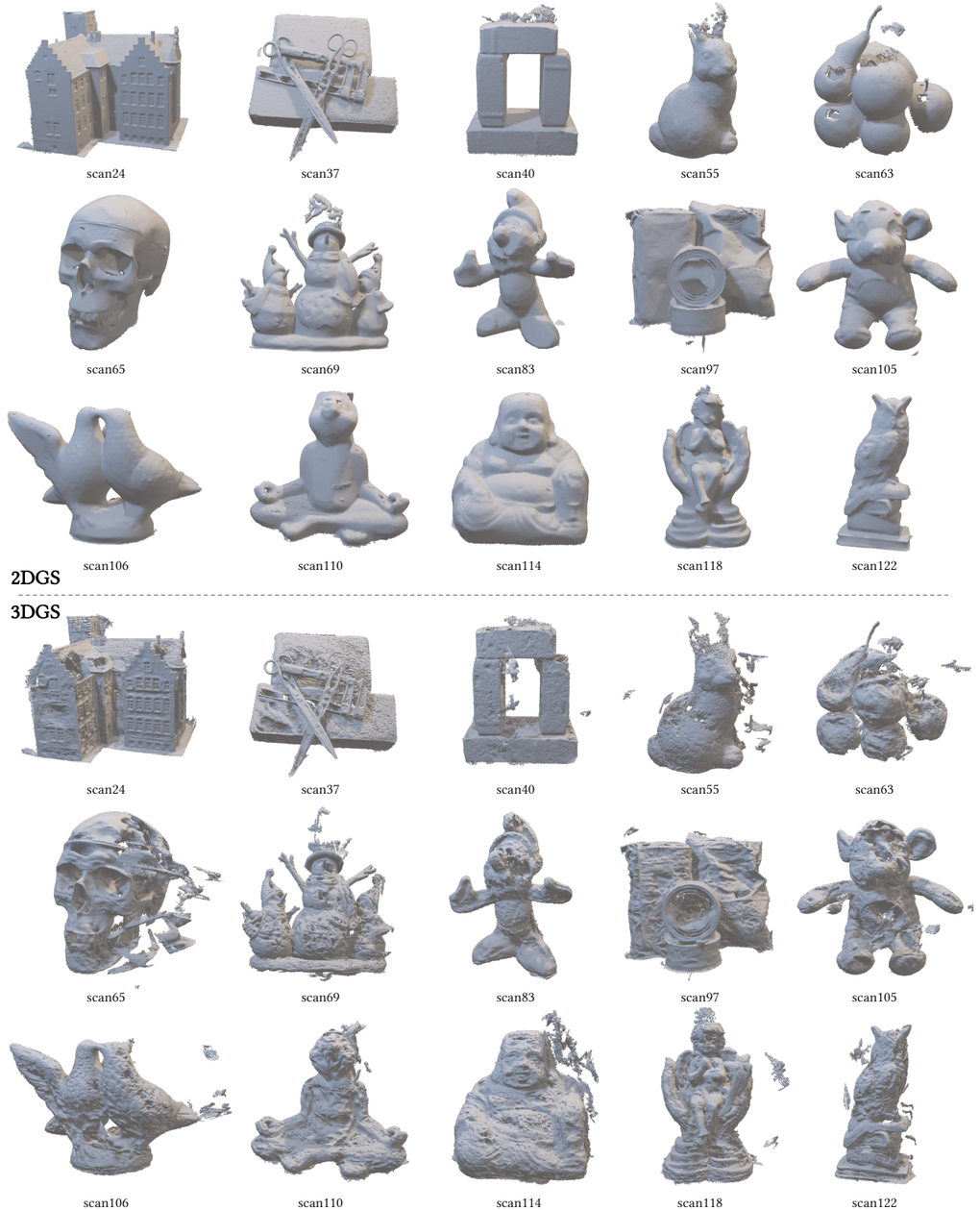


Fig. 9. Comparison of surface reconstruction using our 2DGS and 3DGS [Kerbl et al. 2023]. Meshes are extracted by applying TSDF to the depth maps.

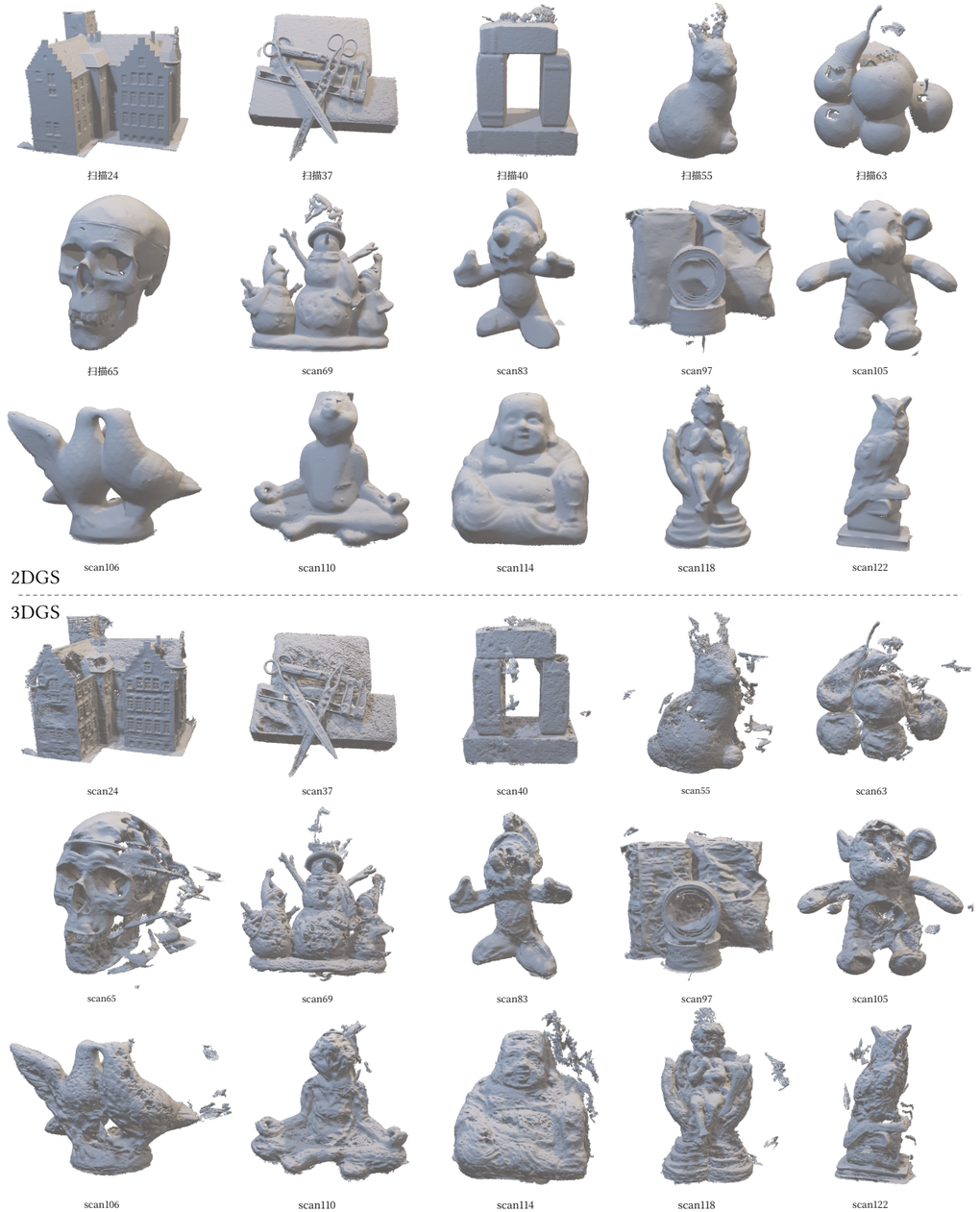


图9. 表面重建的比较　　使用我们的2DGS和3DGS [Kerbl等人 2023]进行表面重建。网格是通过将TSDF应用于深度图提取的　　s.

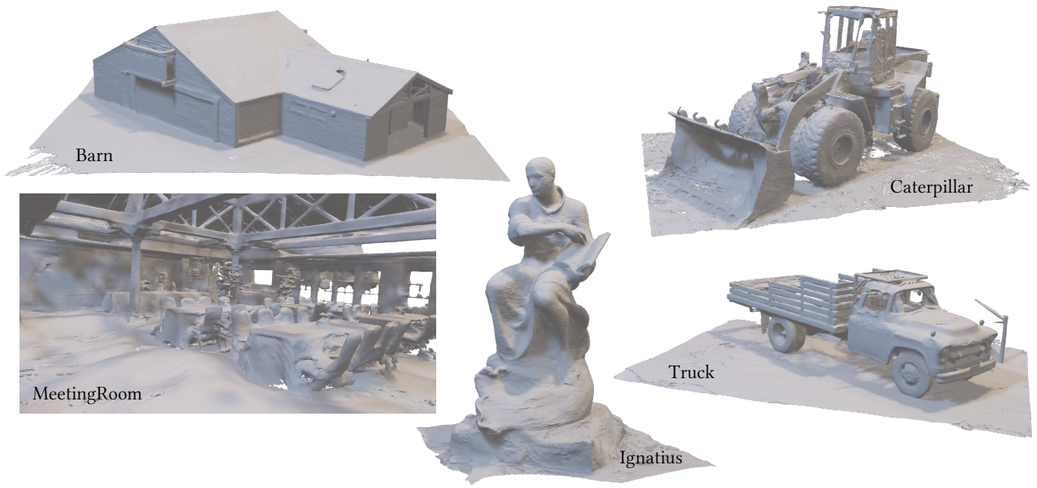


Fig. 10. Qualitative studies for the Tanks and Temples dataset [Knapitsch et al. 2017].

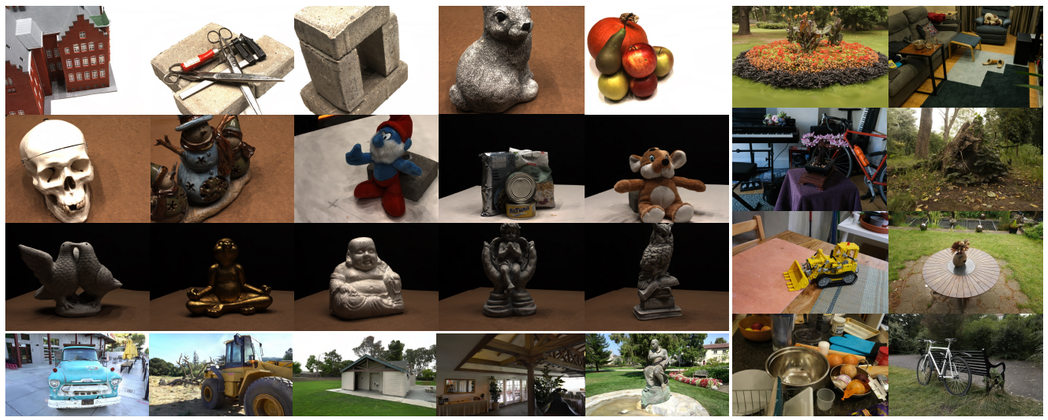
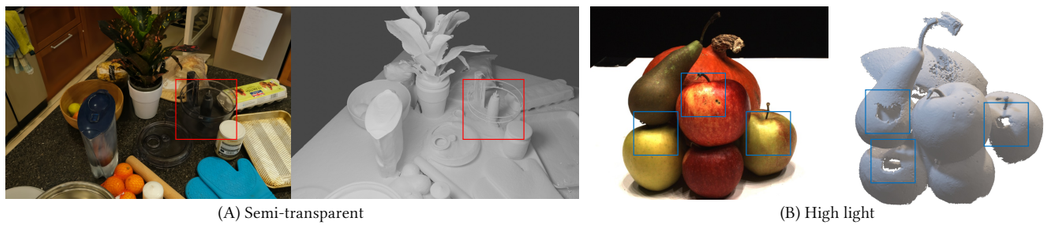


Fig. 11. Appearance rendering results from reconstructed 2D Gaussian disks, including DTU, TnT, and Mip-NeRF360 datasets.



(A) Semi-transparent (B) High light

Fig. 12. Illustration of limitations: Our 2DGS struggles with the accurate reconstruction of semi-transparent surfaces, for example, the glass shown in example (A). Moreover, our method tends to create holes in areas with high light intensity, as shown in (B).

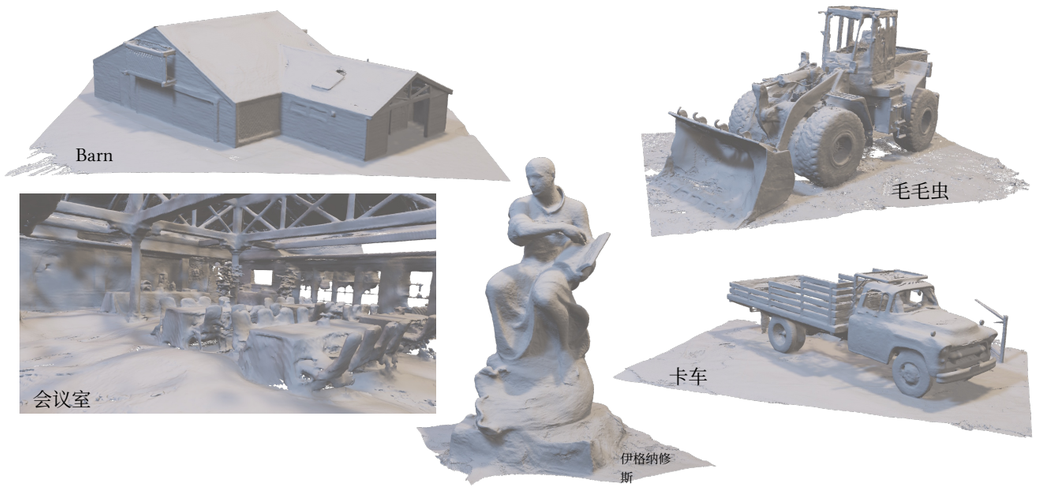


图10。Tanksand Temples数据集的定性研究 [Knapitsch等人 2017]。

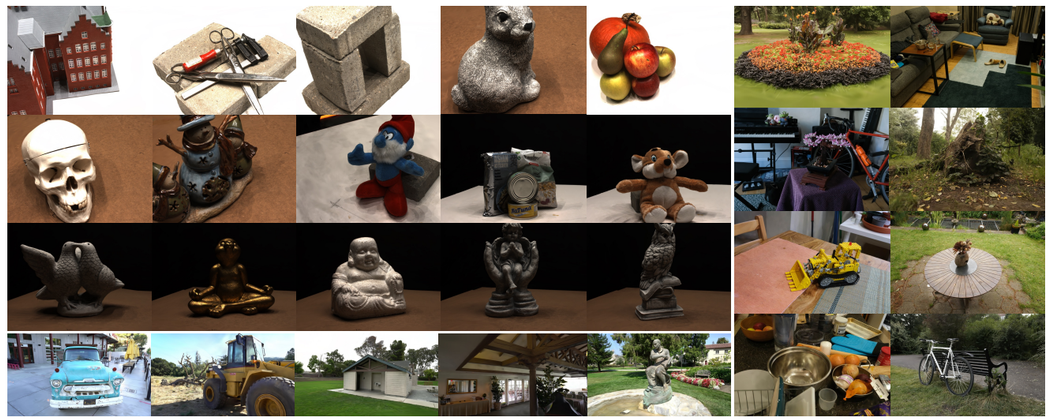
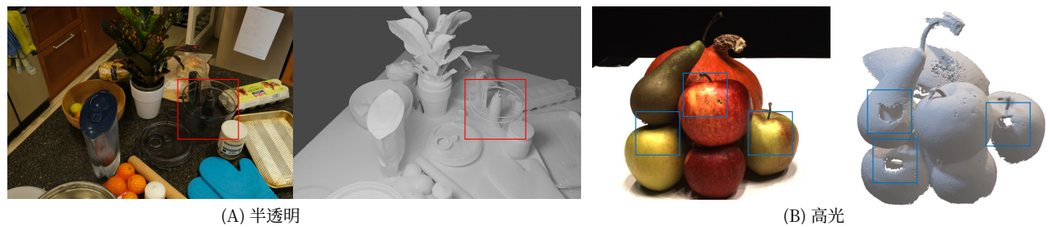


图11。Ap 现渲染结果来自重建的2D高斯圆盘，包括DTU、TnT和Mip-NeRF360数据集。



(A) 半透明 (B) 高光

图12。局限性说明：我们的2DGS难以精确重建半透明表面，例如示例(A)中显示的玻璃。此外，我们的方法倾向于在高光强度区域产生空洞，如(B)所示。